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Advances in facial expression recognition technologies for emotion analysis

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Abstract
With the rapid evolution of technology, automated facial expression recognition (AFER) is one of the emerging research frontiers, gaining increasing attention from computer vision scientists. Facial emotion recognition (FER) enables a wide range of human–machine interaction applications, including mental health monitoring, behavioral analysis, and virtual learning environments. This paper comprehensively reviews recent advancements in AFER, examining both traditional and deep learning (DL) based approaches. The traditional section outlines methods involving feature extraction techniques such as LBP and Gabor filters, along with classifiers like Support Vector Machine (SVM), Principal Component Analysis (PCA), and k-nearest Neighbors (k-NN). In contrast, the DL section explores models such as Convolutional Neural Networks (CNNs) and hybrid networks that offer enhanced accuracy and adaptability. Various public FER datasets, including JAFFE, CK+, and FER2013, are analyzed to demonstrate the scope of existing benchmarks. Furthermore, this study identifies key gaps in the current FER literature, such as limited robustness in real-world conditions, challenges in recognizing emotions, and the lack of standardized evaluation protocols across datasets. We also highlight recent improvements in accuracy, feature extraction, and real-time applicability. This review aims to assist both emerging and established researchers by offering insights into current trends, performance metrics, and unresolved challenges in FER, ultimately contributing to the development of more robust and adaptive emotion recognition systems.

Keywords Automated facial expression recognition (AFER), Human–machine interaction, Traditional FER methodologies, Deep understanding, Feature evaluation, Classification, Research progress in AFER

1 Introduction

Examining facial expressions holds significant importance in conveying emotions non-verbally, aiding the comprehension of individual sentiments and proficiently transmitting their viewpoints, emotions, and feelings. Today, the traditional use of physical documents is gradually being replaced by digital workflows, where laptops or personal computers (PCs) are the primary devices for many tasks. Human–computer interaction has surfaced as a captivating and demanding realm for researchers, covering diverse domains like the analysis of biometric data and the interpretation of human emotional

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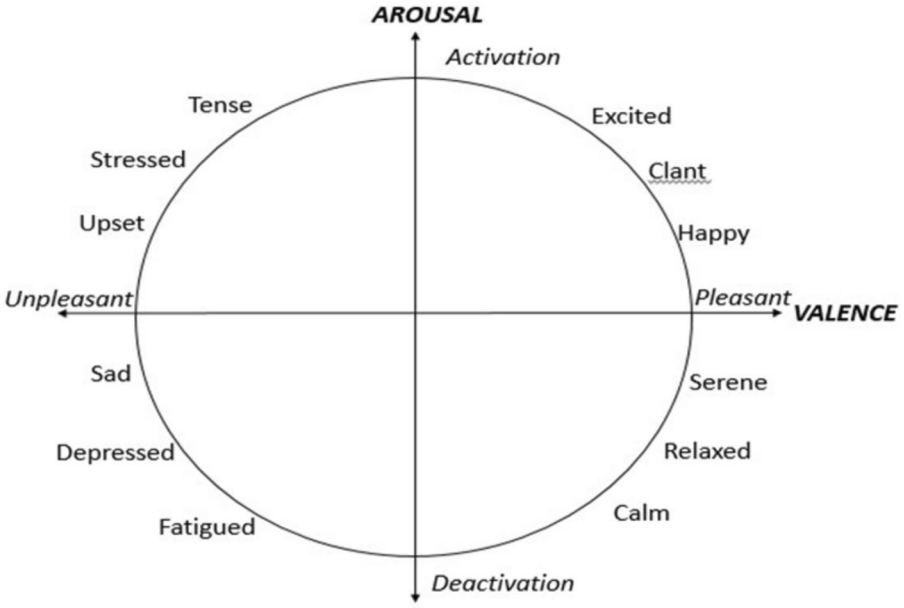


Fig. 1 Representation of emotional dimensions, such as Arousal and Valence expression

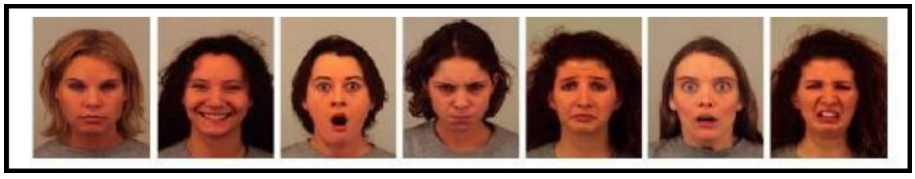


Fig. 2 Seven basic emotions from KDEF database

states [1]. Recognizing human actions relies significantly on emotions, giving rise to various uses across different fields. These applications cover problems like biometric security, evaluating levels of depression or stress, identifying mental conditions, improving video conferencing interactions, aiding online education platforms, assisting forensic inquiries, detecting deception, shaping music according to mood, enabling remote virtual learning, and many more [2].

There exist various methods for categorizing and/or classifying facial expressions. One of the approaches in literature talks about the seven fundamental and universally recognized facial expressions: fear, disgust, sadness, anger, surprise, neutrality and happiness. Each of these is linked to distinct facial action units and referred to as discrete emotional categories. Another approach captures the emotional dimensions delineated within a dimensional framework like Arousal and Valence. Arousal ranges from high to low activation, and Valence spans from unpleasant to pleasant [4], as shown in Fig. 1. The core objective of a Facial Expression Recognition (FER) system is to anticipate or capture these fundamental and supplementary expressions exhibited through a person's facial cues and gestures, subsequently fostering the development of intelligent systems. A visual depiction illustrating the comprehension and categorization of emotions is presented in Fig. 2 [3].

As stated in the background, researchers have explored different methods for expression recognition. One category of approaches is more Conventional, while another type of method employs DL-based techniques.

In the traditional FER approach, a series of fundamental steps are undertaken. The image is initially pre-processed, beginning with an input image, followed by feature extraction using techniques like LBP and Gabor Filter. The image is then classified with machine learning (ML) techniques such as SVM, PCA, k-NN, and Random Forest [1], which leads to the identification of the emotions expressed. In contrast, DL refers to a more advanced and developing area of ML. This approach is usually applied in an FER system, especially when ample amounts of data are available. CNN is a popular DL method use for facial expression recognition. In CNN, images are processed by a set of filters grouped in several convolutional layers to train the database deeply. This procedure produces feature maps which are combined with other feature maps in the fully connected network. At the end of the procedure, the facial expressions are classified through the Softmax function [5].

This technique has three convolutional layers, four pooling layers, three inception layers, and one fully connected layer responsible for recognizing facial features and characteristics. The performance of DNNRL is evaluated on two datasets, FER 2013 and SFEW 2.0. This technique aims to obtain the native mapping of illustrations into Euclidean space to evaluate facial expression likeness concerning relative distance. The precision of this method is tested against three other networks: Alexnet, DNN, and T-DNN, where DNNRL is found to be the best performer.

Dapeng Tao [74] provided a new facial recognition algorithm, Tensor Rank Preserving Discriminant Analysis (TRPDA), which entails a two-step process. The first step is obtaining a lower-dimensional tensor subspace from the input sensor arrays. The second step consists of applying a discriminative locality alignment to obtain the vector feature for frontal face recognition. Anticipating real-time fluctuations in human emotional responses through analysing a vast array of physiological signals originating from internal bodily organs has been investigated in a work referred to in [73]. This introduces a methodology known as reinforcement online learning (ROL) to estimate the emotional states. This approach harnesses rewards to adapt the predictor throughout successive online training iterations. During each iteration, rewards are estimated, which guides the selection of specific instances for predictor refinement. The ROL technique is utilized to optimize least-squares (LS) and employs support vector regression (SVR) to predict emotions. The performance of this approach showcases efficient learning times and yields robust prediction models within minimal time. The practical viability of this approach is validated through its application to both Artificial and DEAP datasets.

Familiarity with fundamental terminology is essential before delving into the various approaches of the FER system, aiding the comprehension of the FER system concept.

1.1 Knowledgeable facts

Human emotions analysis using facial expressions involves the study of emotions by analyzing the subtle changes and patterns in a person's facial features. This field, also known as FER, plays a significant role in various applications such as psychology, human–computer interaction, marketing, and more. Here are some discovered facts about human emotion analysis using facial expressions:



Fig. 3 Examples of compound emotional states include delight mixed with surprise, regret combined with surprise, anger tinged with surprise, terror accompanied by surprise, disgust intertwined with surprise, and pure surprise

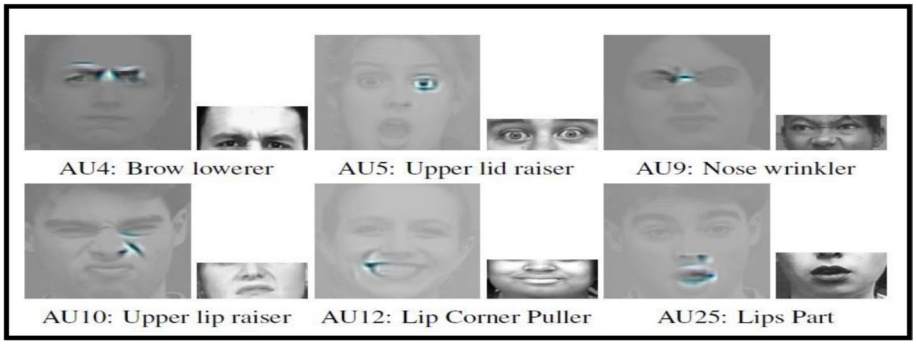


Fig. 4 Different action units

- In the nineteenth century, significant progress was made in facial expression analysis, particularly focusing on six core emotions, i.e. joy, sadness, fear, disgust, surprise, and anger. In 1978, American psychologists Ekman and Friesen introduced the Facial Action Coding System (FACS) [66]. It is well illustrated in Fig. 2.
- The combination of two fundamental emotions is called a compound emotion. Du [7] presented 22 emotions, of which seven are basic emotions, 12 mixed emotions and three complementary emotions, as shown in Fig. 3.
- The FACS system starts by detecting the face, then proceeds to segment the face into different regions defined as action units (AU), and later, it can track face movements. This relies on the contracting of facial muscles, which tends to correspond to the emotional states of a person [8, 13], as can be seen in Fig. 4.
- FACS comprises 46 fundamental action units defining the transitions on the face due to actions and even speech. It captures expressions unique by action, like raising the inner eyebrow, wrinkling the nose, and moving lips. Undoubtedly, it is easier to express emotions using FACS by observing single action units or multi-united sights where a variety of action units must be intermingled together for sensible expressions [9]. For instance, the algorithm could recognise the raised outer eyebrows as AU2 and a fearful expression would be tracked. The set of AUs $(1+2+5+10)$ captures the facial expression of surprise accompanied by disgust, as indicated in Table 1.
- Part of the facial features are the tip of the eyebrow, the nose's base, the eye's angle, and the mouth. These features elicited from the photograph serve as outputs to be processed by the DL model. The DL methods can be subdivided into three segments, one of which is facial landmark feature extraction. These are (i) Active Shape Models (ASM) and Appearance Models (AM), (ii) Regression Models, and (iii) Convolutional Neural Network (CNN) Models [5].

Table 1 Action Units of basic and compound emotions

Emotions	Action Context	Emotions	Action Context
Joy	[12, 25]	Sad	[4, 15]
Sadly-Disgusted	[4, 10]	Fear	[1, 4, 20, 25]
Fear with Surprise	[1, 2, 5, 20, 25]	Fear with Angerness	[4, 20, 25]
Fear with disgusted	[1, 4, 10, 20, 25]	Surprised	[1, 2, 25, 26]
Angry	[4, 7, 24]	Hatred	[4, 7, 10]
Sadly-Angry	[4, 15]	Happily-Sad	[4, 6, 12, 25]
Disgusted	[9, 10, 17]	Happily-Fearful	[1, 2, 12, 25, 26]
Awed	[1, 2, 5, 25]	Happily-Disgusted	[10, 12, 25]
Happily-Surprised	[1, 2, 12, 25]	Sadly-Fearful	[1, 4, 15, 25]
Angrily-Disgusted	[4, 10, 17]	Angrily-Surprise	[4, 25, 26]
Disgusted-Surprise	[1, 2, 5, 10]	Appalled	[4, 9, 10]
Sadly-Surprise	[1, 4, 25, 26]		

- The application of FER is split into two categories, static images in the first and sequences of images in video clips in the second [10]. The latter is obtained through the evaluation of image features by point- or texture-based methods. M. Pantic [11] proposed a point-based model that consists of two two-dimensional views, frontal and profile. The frontal view model captures thirty features, out of which 19 facial landmarks define 25, while the remaining are some preset contour shapes of the chin and the mouth. The side-profile face framework utilizes ten facial landmarks corresponding to the maximum and minimum values of the face's curvature. As stated by Edwards [12], Texture-Based Methods mainly use active appearance models with facial images on which 122 landmarks around the facial components are used to build the model. These features are derived through three techniques, which are optical flow, feature point tracking, and 3d modelling.

1.2 Evolution of facial expression recognition techniques

Facial Expression Recognition (FER) has witnessed a remarkable transformation over the past two decades, progressing from rule-based and handcrafted approaches to data-driven and context-aware DL models. Figure 5 presents a chronological timeline outlining significant milestones in FER research from 2005 to 2025. This evolution reflects methodological innovations and a shift in design goals ranging from achieving basic accuracy in controlled settings to enabling robust, real-time, and explainable emotion recognition in unconstrained environments. In the early stages (2005–2012), FER systems predominantly relied on traditional ML techniques, including SVMs, KNN, and Decision Trees, combined with handcrafted feature extraction methods such as Local Binary Patterns (LBP), Haar features, and PCA. These systems are typically evaluated on small, posed datasets such as JAFFE, CK+, and MMI. While these techniques laid the foundation for FER, their performance is limited by sensitivity to illumination, pose variation, and lack of generalization to real-world scenarios. The period from 2013 onward marks a paradigm shift with the adoption of DL, particularly CNNs, which enabled automatic feature learning from raw pixel data. CNNs proved highly effective in capturing spatial hierarchies in facial images, resulting in significant performance improvements on benchmarks like FER-2013, RAF-DB, and AffectNet. Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks further extended FER to video sequences, enabling the modeling of temporal dynamics. This era also introduced hybrid

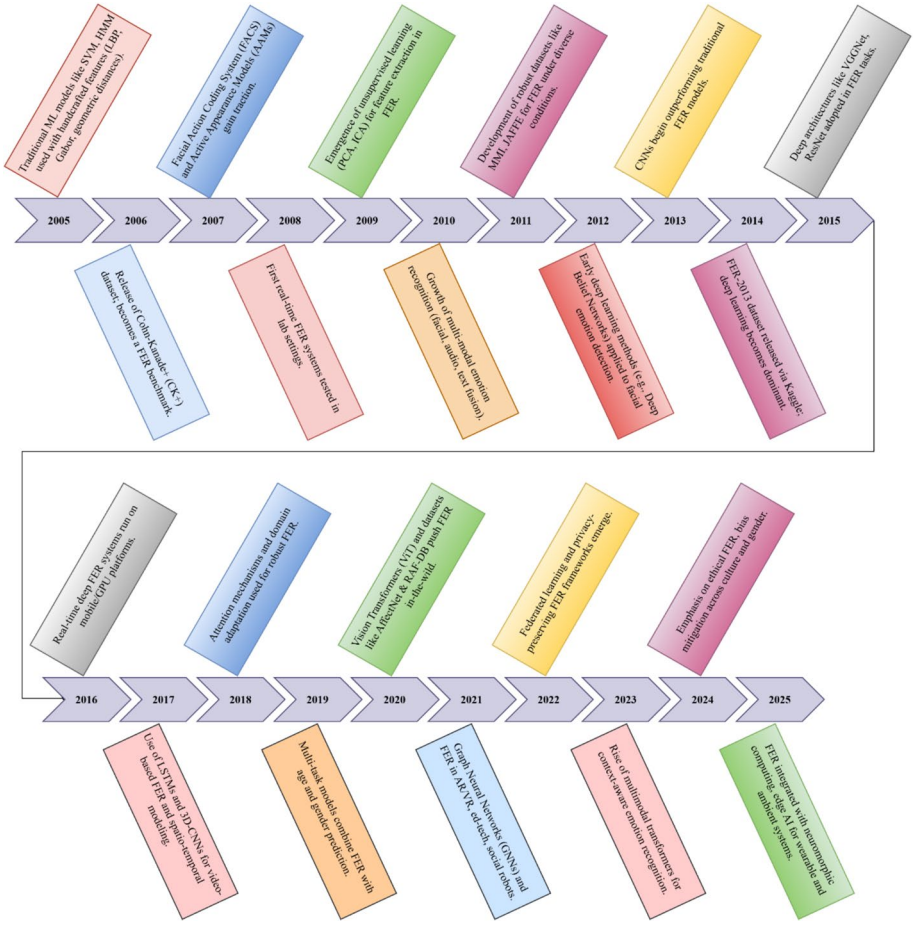


Fig. 5 Evolution timeline of Facial Expression Recognition (FER) methods and milestones from 2005 to 2025

models combining CNNs with RNNs, attention mechanisms, and ensemble learning to improve accuracy and robustness.

In recent years (2019–2025), FER research has increasingly focused on addressing deployment challenges and interpretability. Lightweight architectures such as MobileNet and EfficientNet have been explored for on-device FER in mobile and edge environments. Vision Transformers (ViTs), Graph Neural Networks (GNNs), and Explainable AI (XAI) frameworks have opened new directions by enabling global reasoning, structural representation, and transparency in emotion inference. Simultaneously, multimodal FER, incorporating audio, video, depth, and physiological signals, has emerged as a promising solution to overcome the limitations of unimodal systems. Current trends emphasize personalization, fairness, and ethical deployment, especially in sensitive applications like healthcare, education, and surveillance.

Thus, the evolution of FER has not been linear but multidimensional, driven by advances in ML, the availability of large annotated datasets, and the increasing demand for emotionally intelligent systems. The next generation of FER systems is expected to be more adaptable, transparent, and human-centric, bridging the gap between academic development and practical deployment.

1.3 Our contribution

This paper aims to comprehensively survey both traditional and DL-based FER approaches, critically examining their methodologies, strengths, and limitations. The core objectives are to (i) review the evolution of FER techniques and their comparative performance, (ii) identify current research gaps such as real-world robustness and dataset standardization, and (iii) analyze how benchmark datasets influence model development and evaluation.

To guide this exploration, we consider the following research questions:

- How have traditional and DL methods evolved in FER, and what are their respective advantages and limitations?
- What are the major challenges in achieving reliable FER in unconstrained, real-world environments?
- How do FER models perform across different datasets, and what metrics best capture their effectiveness?

In addition to theoretical advances, this paper also highlights the practical integration of FER in various industries. For instance, in healthcare, FER assists in diagnosing mental health conditions by analyzing patients' emotional cues. In education, it enhances virtual learning by monitoring student engagement and emotional states. In the security sector, FER is employed in surveillance systems to detect suspicious behaviors in public spaces. These examples underscore the growing relevance and impact of FER technologies across domains.

This survey is based on research articles published between 1978 and 2025, categorized into six time periods (as shown in Fig. 6). A significant majority of the research, accounting for 60%, has been published between 1978 and 2000, indicating a strong historical foundation in the literature. The following periods saw a noticeable decrease in contributions: 2% (2001–2005), 8% (2006–2010), and 11% (2011–2015). However, a resurgence of interest has been observed in more recent years, with 19% of articles published between 2016 and 2020 and 34% from 2021 to 2025, though the latter might

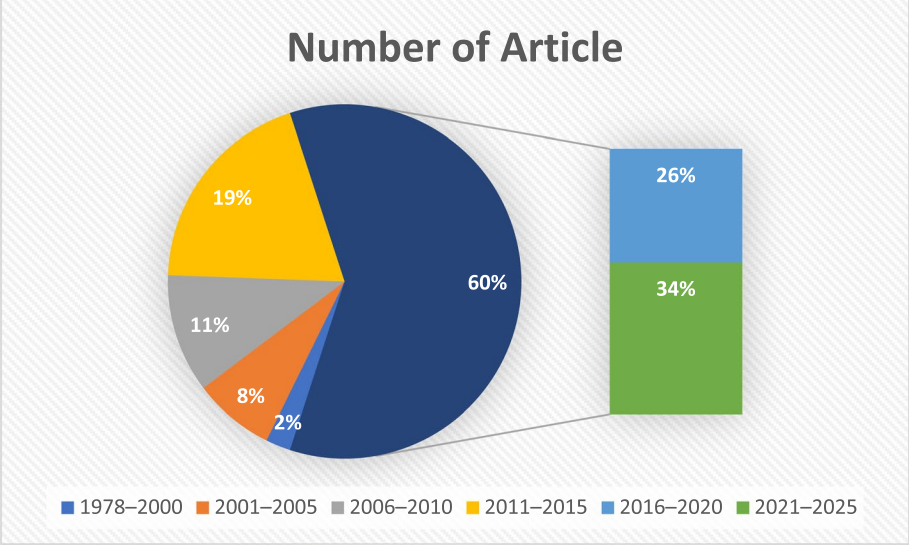


Fig. 6 Temporal Breakdown of Research Articles Used in the Survey

include projected or in-press publications. This distribution highlights the evolving interest and growing research output in the field over time.

1.4 Manuscript's structure

The structure of this manuscript is outlined as follows. Section 2 presents conventional approaches, including a compilation of research summaries on this approach. The introduction of DL, related research, and hybrid models as novel techniques in this field are also elaborated. Sections 3 and 4 delve into the discussions about standard public databases for the FER system and challenges and performance measures in facial expression recognition. At last, Sect. 5 concludes by addressing benchmarks achieved and prospects for future work in this field.

2 Classification of facial expression approaches

The classification of methods for analyzing facial expressions involves categorizing various strategies, techniques, and methodologies employed to identify and understand facial expressions. These approaches encompass a range of methods, algorithms, and models utilized to interpret emotional states based on facial cues. The categorization facilitates comprehension of the diverse techniques available for investigating facial expressions, spanning domains such as psychology, computer vision, and human–computer interaction. The classification is shown in Fig. 7 and is discussed as follows.

- *Conventional approach* This method aligns with established practices in emotion classification. It is most effective with smaller datasets, even though including modern feature extraction algorithms and classification techniques can enhance its performance.
- *Deep learning approach* This contemporary strategy is employed to classify emotions and is particularly suitable for extensive datasets. Various forms of CNN and Recurrent Neural Networks (RNN) can be employed to train and test the images. Presently, the ResNet network is prevalent across various applications.
- *Hybrid approach* This approach increases precision and effectiveness by merging conventional feature extraction methods and DL models. It offers greater performance improvement when tackling complex datasets and real-world problems because it utilizes the advantages of both methodologies.

2.1 Conventional approach

Different studies using traditional techniques, deep-learning approaches, and hybrid techniques have researched facial expression recognition [77, 87–90, 93]. The conventional technique, however, continues to be one of the most researched techniques because it works well under resource-constrained environments. The basic contour of

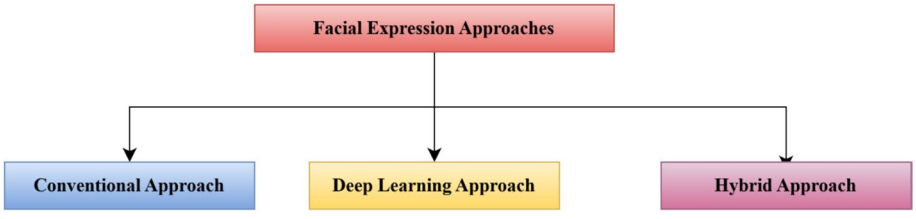


Fig. 7 Classification of FER approach

the processes of emotion recognition is given in Fig. 8. The process of traditional FER techniques involves five distinct stages: (a) Preprocessing phase, (b) Facial Detection, (c) Identification of facial components such as mouth, lips, and eyes, (d) Utilization of Face Extraction techniques, and finally, (e) Application of Classifier techniques, leading to the eventual classification of emotions.

2.1.1 Image pre-processing

Image pre-processing [94, 96, 97, 104, 108] is an elementary stage applied to raw images to eliminate redundant noise while preserving valuable information. This process is designed to mitigate potential alterations in the image caused by environmental and illumination variations if they are present [14]. In some cases, real-time images may suffer from distortion and noise, necessitating preliminary filtering using techniques such as the Gabor filter, Mean, Median filter, and Sobel filters [15]. Such filters are employed to improve image smoothness without compromising essential information.

In scenarios where the database size is not standardized, certain methods can be employed to achieve an optimal image size for subsequent processing, thereby enhancing system performance [71]. Certain methods can be used in the image pre-processing step, as discussed below.

- *Synthetic Sample Generation* When the available image data is insufficient to meet the application's performance requirements, augmenting the database with authentic images through artificial rotations, translations, and skewing can be employed, often by combining two or more transformations for improved results.
- *Rotation Correction* Images of the same subject can exhibit rotation, size, and brightness variations. Aligning the facial region with the horizon and center point addresses these issues, ensuring correct images for further rotations and translations.
- *Down Sampling* Utilizing linear interpolation, this operation minimizes image dimensions for the network while simultaneously achieving scale normalization, ensuring consistent positioning of all facial components.
- *Intensity Normalization* To counter variations in brightness and contrast within the same image, intensity normalization is employed, mitigating the complexity arising from these fluctuations.
- *Image Cropping* To eliminate background elements that could reduce classification accuracy, background and foreground differentiation is achieved through cropping, retaining only the essential parts of the image.

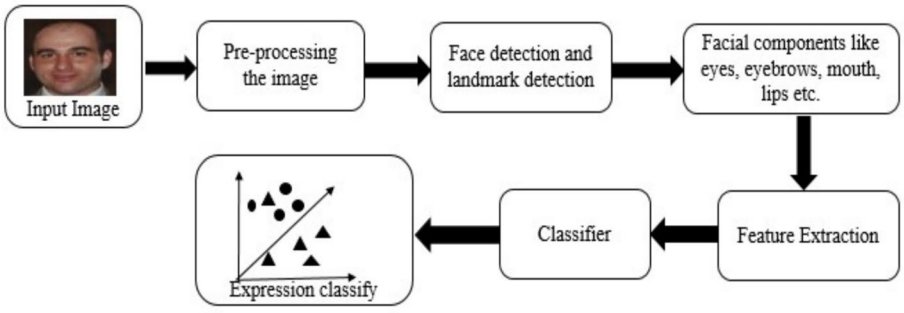


Fig. 8 Design Flow of Traditional FER

Kulkarni [16] used the Weighted Least Square (WLS) method to accommodate disparities across age groups and comprehend human emotions. By eliminating wrinkles on the face, this technique counteracts the impact of aging. The grayscale picture 't' used in the WLS technique is divided into two layers: the detail layer ('d') and the base layer ('b').

To decrease the expression of an energy function, the equation is outlined as follows:

$$\sum \left((b_p - t_p)^2 + \mu \left(\alpha_{x,p}(t) \left(\frac{\partial b}{\partial x} \right)_p^2 + \alpha_{y,p}(t) \left(\frac{\partial b}{\partial y} \right)_p^2 \right) \right) \tag{1}$$

where μ = Adjusting coefficients to optimize the value of μ results in smoother base layer b, ay = smoothness weights in the y-direction, axe = smoothness weights in the x-direction, p = pixel's spatial position.

$$a_{x,p}(t) = \left(\left(\frac{\partial t}{\partial x}(p) \right)^a + \varepsilon \right)^{-1}, a_{y,p}(t) = \left(\left(\frac{\partial t}{\partial y}(p) \right)^a + \varepsilon \right)^{-1} \tag{2}$$

2.1.2 Face and Landmark detection

Detecting the face involves processes such as isolating the face from the entire image, identifying facial components like eyes, mouth, lips, and nose, and detecting landmarks within the facial area while excluding other objects like trees or buildings. The image is segmented to extract the face or landmarks' components, even under varied illumination and unclear surroundings. This complexity contributes to challenges in emotion identification. Researchers employ multiple face detection algorithms to recognize the human face within an image efficiently. Such techniques include Viola-Jones [17], Branch and Bound [18], Exhaustive Search [18], and skin color modelling [78]. The Haar Cascade technique is employed to identify components like eyes and mouth in the face area, utilizing a ML approach dependent on extensive positive and negative image training cascade functions. However, challenges emerge when detecting faces due to variations in poses, motion, image orientation, lighting conditions, occlusions, background colour, texture discrepancies between faces, age-related differences, and noisy and blurry images, among other factors [178]. These challenges arise during image acquisition and can impact the system's efficacy.

In the face detection process, an image or video is made input to face detection algorithms to locate the face simultaneously. While some systems first detect the face and then locate it, as illustrated in Fig. 9. Traditional face detection algorithms mainly focus on frontal 2D images. Yet, newer algorithms can be employed for multi-dimensional 3D images, outlining landmarks within the face [17]. The face detection algorithm is detailed in Table 2.

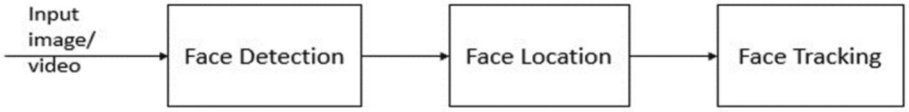


Fig. 9 Steps of Face Detection

Table 2 Traditional methods based on face and landmark detection

Method of Detection	Explanation	Advantage	Limitations	Resuls
Viola-Jones Method [17]	This approach calculates the cumulative sum of picture pixels inside a rectangle using Haar-basis functions	The real-time challenge of face detection is effectively addressed, along with proficient feature selection	This method is not suitable for posed variation faces under varying lighting conditions captured images	~94% accuracy
Exhaustive Search Method [19]	Drawing from evolutionary agent theory, a window search method is employed to detect faces in gray-scale images	Evaluate the potential feature sets from the face trajectories. This approach is considered optimal	Computation is complex	~89% accuracy
Branch and Bound Method [177]	Haar wavelet features, added with the ADA Boost M1 algorithm, are used for facial recognition to minimize computational costs	This method gives the optimal result for Branch and Bound algorithm	Time-consuming method	~92 accuracy in face recognition
Skin color modelling [78]	The purpose is to identify skin color pixels from non-skin color pixels, aiding in the elimination of skin-colored backgrounds. Pixels with an intensity of 1 are categorized as skin pixels, while others are classified as non-skin pixels	Various color models and classifiers are employed to enhance the detection rate	The system may fail to detect faces due to misleading background colors	Scored 65%-90% accuracy in detecting face evaluated over different color space

2.1.3 Feature extraction & reduction methods

Every image has certain metrics associated with it known as features. Such features are obtained from the system's image using different techniques within the facial expression system. Image processing requires substantial computing power, and imaging feature extraction processes help in reducing that time by restructuring the image/video data to capture more meaningful forms. Many techniques are used for feature extraction in FER including Local versus Holistic approaches, Action Unit versus Non Action Unit distinctions, Geometric-based methods, and Appearance-based methods as well as hybrid methods combining both Geometric and Appearance features.

- Holistic vs Local approach:

Facial expression recognition is a field within computer vision and artificial intelligence that focuses on interpreting human facial expressions through various technologies. The holistic approach considers the entire face as a single entity when recognizing facial expressions. It analyzes the global facial features and their spatial relationships without explicitly focusing on individual facial components. Neural networks [8] and Gabor Wavelets [18] are popular holistic approaches. Conversely, local-based strategies concentrate on particular facial regions sensitive to emotional changes. It involves identifying key facial landmarks, such as eyes, nose, and mouth, and analyzing their changes to recognize emotions. The choice is dependent on the task; holistic is good for overall context, while local excels in detecting fine-grained patterns [19, 20].

- Geometric vs Appearance-based approach:

Geometric techniques are based on finding out about the positions and movements of landmarks like eyes, nose and mouth on the face. They usually keep track of the movements of these landmarks to see how facial expressions change over time. Here,

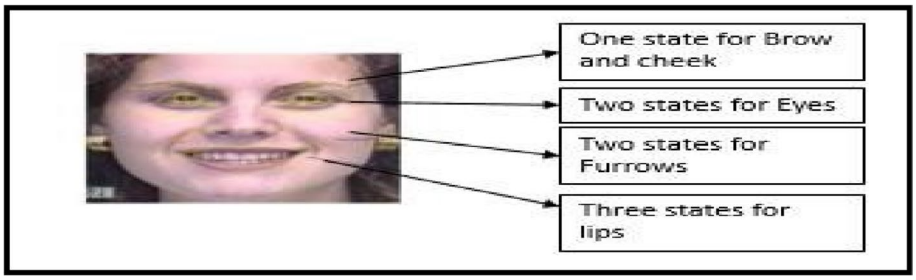


Fig. 10 Geometry-based feature extraction method

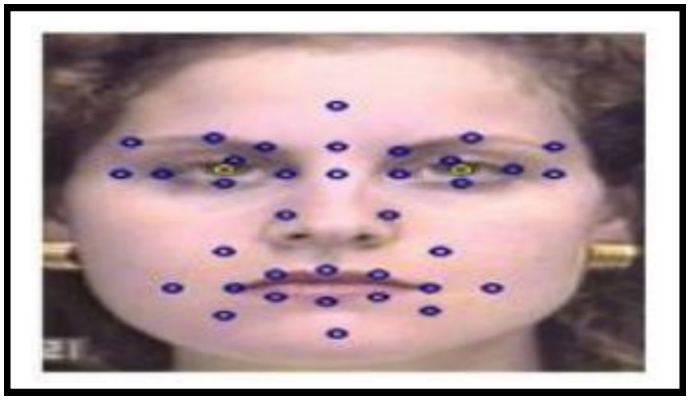


Fig. 11 Appearance-based features Extraction approach

landmarks or facial points are applied in Geometric or Intransient-based attributes to make the feature vector as explained and shown in Fig. 10 [20]. As outlined in [21], two feature types are deployed, contingent upon the positioning and orientation of 52 facial landmark points. The initial approach entails the computation of angles and Euclidean distances between pairs of facial landmarks, including the nose, eyes, ears, and mouth. Alternatively, the second method involves deriving distances and angles by subtracting them from their corresponding values in the initial frame.

Wrinkles or creases that appear near the eyes, nose and mouth, influenced by facial expression and shown in Fig. 11 [19], are said to be appearance or transient features. These attributes entail employing image filters across the entire image or designated facial regions to derive the feature vector. Some of the techniques used to get image features are Local Binary Patterns (LBP), Gabor Wavelet and filtering, Local Directional Patterns (LDP), Local Directional Ternary Patterns (LDTP) and Edge of Histogram (EOH) [14].

In order to improve results in specific situations, many researchers have implemented a blending or amalgamation approach that combines both geometric and appearance-based features [22], effectively overcoming limitations associated with individual features.

2.1.3.1 Feature extraction methods

Several techniques for extracting features exist, including Gabor filters, Log Gabor filters, Local Binary Pattern (LBP), Local Gradient Coding (LGC), Local Directional

Pattern (LDP), Histogram of Oriented Gradients (HOG), and Scale-Invariant Feature Transform (SIFT), among others.

- *Gabor Wavelet* This technique for extracting features employs a global spatial-oriented strategy, encompassing the entire image rather than focusing on smaller image regions. It serves a dual purpose, serving both as a preprocessing step for the image and as a means to extract features. The initial concept is introduced by Gabor [67], functioning across spatial and frequency domains to achieve optimal attributes. With this technique, even and odd filters are bandpass filters and are multiplied as shown in Eqs. 3 and 4.

$$g_e(x) = \frac{1}{\sqrt{2\pi\sigma}} e^{-\frac{x^2}{2\sigma^2}} \cos(2\pi\omega_0 x) \tag{3}$$

$$g_0(x) = \frac{1}{\sqrt{2\pi\sigma}} e^{-\frac{x^2}{2\sigma^2}} sin(2\pi\omega_0 x) \tag{4}$$

where,

- ω_0 = central frequency at which the filter produces its peak response.
- σ = spread of the Gaussian window.
- *Histogram of Oriented Gradient (HOG)* [75, 76] HOG proves to be an immensely effective technique for feature extraction from facial expressive images. Various HOG variants are employed across diverse applications, yielding optimized outcomes. Functioning as a shape descriptor, HOG characterizes object shapes through edge directions. It also assesses the prevalence of gradient orientations within localized image segments. It calculates gradient orientations and magnitudes by applying the gradient operator to the image. This descriptor is founded on amalgamations of gradient directions across pixels within small spatial regions, referred to as 'cells'. Subsequently, a 1D histogram is formed, followed by concatenation to create the feature vector, which serves as the basis for subsequent stages. Assuming 'I' is a grayscale (intensity) function representing the image under analysis, the image is divided into cells measuring N*N pixels, with gradient orientation $\theta_{x,y}$ computed for each pixel. This process is detailed in Eq. (5).

$$\tan^{-1} = I(x, y+1) - I(x, y-1) / I(x+1, y) - I(x-1, y) \tag{5}$$

Subsequently, the orientations θ_i^j in which i ranges from 1.....N^2 residing to the same cell j are quantized and collected into an M- bins histogram. Lastly, all formed histograms are ordered and formulated into a unique HOG histogram that is the final result of this descriptor and its outcome is used for the next processing.

- *Edge Oriented Histogram (EOH)* This feature uses a local spatial technique to produce a simpler version of the histogram of gradients (HOG). According to [23] [24], EOH is appropriate for facial recognition and object detection. EOH is ideally suited for determining the gradients of borders in various directions when it comes to detecting the corners and borders of a face, and this procedure also requires less time to complete [23]. First, the input picture is split into blocks I(x,y) and transformed into a grayscale image. Equation (6) illustrates how the Sobel edge operators (SX and

SY) [1] are used to assess the horizontal (Fx) and vertical (Fy) edges for each block. The EOH feature vector is formed by splitting the angle (θ) and strength S(x,y) of the various edge orientations into N bins, as indicated in Eqs. (7) and (8).

$$\mathrm{F_x}\left(\mathrm{x},\mathrm{y}\right)=\mathrm{S_x I}\left(\mathrm{x},\mathrm{y}\right) and \mathrm{F_y}\left(\mathrm{x},\mathrm{y}\right)=\mathrm{S_y I}\left(\mathrm{x},\mathrm{y}\right) \tag{6}$$

$$\mathrm{Angle}\left(\theta\right)=\arctan\left(\frac{\mathrm{G_y}\left(\mathrm{x},\mathrm{y}\right)}{\mathrm{G_y}\left(\mathrm{x},\mathrm{y}\right)}\right) \tag{7}$$

$$\mathrm{S}\left(\mathrm{x},\mathrm{y}\right)=\sqrt{\mathrm{G_x^2}\left(\mathrm{x},\mathrm{y}\right)+\mathrm{G_y^2}\left(\mathrm{x},\mathrm{y}\right)} \tag{8}$$

- *Local Binary Pattern (LBP)* Additionally, this feature is calculated using an image's local feature [25]. This method works well for applied operations on grayscale pictures and texture analysis [26]. The facial expression recognition system published in [27] exemplifies how the LBP approach is applied in face detection. Each pixel's LBP is calculated using the pixels in its neighbourhood. Equation 9 may be explained as follows: each surrounding pixel mi has the threshold value removed, the number of surrounding pixels is represented by pi and the centre pixel is represented by pc.

$$\mathrm{LBP_i}=\sum_{\mathbf{i}=0}^{\mathrm{m}-1}\mathrm{f}\left(\mathrm{p_i}-\mathrm{p_c}\right).2^{\mathrm{i}}\ \mathrm{f}\left(\mathrm{x}\right)=\begin{cases}1x\geq 0\\ 0x<0\end{cases} \tag{9}$$

Multiple variants of LBP are used to optimize the feature vectors like Local Ternary Pattern (LTP), Completed local binary pattern (CLBP), Extended local binary pattern (ELBP), Multi-quantized local binary pattern (MQLBP), Local Gradient Code (LGC) and all these are used for special purpose [79]. Here, we discuss only the LGC feature descriptor.

- *Local Gradient Code (LGC)* LGC [28] is dependent on the relationship with the neighboring pixel value, whereas LBP equates the center pixel value with the surrounding pixel value. Two versions of the optimized LGC exist. Equation (12) [2 describes the second, which is based on the principle of vertical diagonal LGC-VD and the first, which is dependent on the principle of horizontal diagonal LGC-HD (11).

$$\begin{aligned}\mathrm{LGC}=&\mathrm{s}\left(\mathrm{g_1}-\mathrm{g_3}\right)2^7+\mathrm{s}\left(\mathrm{g_4}-\mathrm{g_5}\right)2^6+\mathrm{s}\left(\mathrm{g_6}-\mathrm{g_8}\right)2^5+\mathrm{s}\left(\mathrm{g_1}-\mathrm{g_6}\right)2^4\\&+\mathrm{s}\left(\mathrm{g_2}-\mathrm{g_7}\right)2^3+\mathrm{s}\left(\mathrm{g_3}-\mathrm{g_8}\right)2^2+\mathrm{s}\left(\mathrm{g_1}-\mathrm{g_8}\right)2^1+\mathrm{s}\left(\mathrm{g_3}-\mathrm{g_6}\right)2^0\end{aligned} \tag{10}$$

$$LGC-\mathrm{HD_P^R}=\mathrm{s}\left(\mathrm{g_1}-\mathrm{g_3}\right)2^4+\mathrm{s}\left(\mathrm{g_4}-\mathrm{g_5}\right)2^3+\mathrm{s}\left(\mathrm{g_6}-\mathrm{g_8}\right)2^2+\mathrm{s}\left(\mathrm{g_1}-\mathrm{g_8}\right)2^1+\mathrm{s}\left(\mathrm{g_3}-\mathrm{g_6}\right)2^0 \tag{11}$$

$$\mathrm{LGC}-\mathrm{VD_P^R}=\mathrm{s}\left(\mathrm{g_1}-\mathrm{g_6}\right)2^4+\mathrm{s}\left(\mathrm{g_2}-\mathrm{g_7}\right)2^3+\mathrm{s}\left(\mathrm{g_3}-\mathrm{g_8}\right)2^2+\mathrm{s}\left(\mathrm{g_1}-\mathrm{g_8}\right)2^1+\mathrm{s}\left(\mathrm{g_3}-\mathrm{g_6}\right)2^0 \tag{12}$$

2.1.3.2 Feature reduction To handle the difficulty of managing wide feature vectors and finding useful patterns in large data, certain methods exist to reduce dimensionality without omitting important data. Among the methods we can use are PCA [28], methods to deal with many missing values, algorithms to lower the effect of minor variances, methods like Sammon's mapping [29] or ISOMAP, Backward or Forward Feature Elimination, initial steps in random forest selection and Linear Discriminant Analysis (LDA) [155–157].

- *Principal Component Analysis (PCA)* [1, 13] is a statistical method used for reducing dimensionality. It is applied before the ML phase to compress data. The process of PCA-based data reduction consists of several steps. Let the dataset include p observations of an n-dimensional vector x, each element of which is $\{x_1, x_2, x_3 \ldots X_n\}^T$. The mean of the vector should be zero, so take the mean of each observation p, subtract the mean βx of X from each component, as shown in Eq. (13). Researchers have effectively utilized this approach to yield favourable outcomes.

$$\beta_X = \frac{1}{n}\sum_{i=1}^{n}(x_i) \tag{13}$$

$$S_{\mathbf{x}} = \sum_{i=1}^{n}(x_i - \beta_i)(x_i - \beta_i)^T \tag{14}$$

The covariance matrix S_x is calculated to get the eigenvalues and eigenvectors. The method of obtaining relevant picture transition is to provide less information about the image to describe high and low variance. This means, according to equation 14, keeping the primary high-order components and not using the lower-order components [12].

- *Sammon Map* [29] A non-linear feature extraction/reduction technique for 2-D representation of high dimensional data is the "Sammon map" [29]. To map from D-TO-D dimensions, Summon's approach uses a gradient descent technique to reduce an error function. It applies to all dimensionalities when d < D. The error function can be shown as:

$$E = \left(1 \Big/ \sum_{J=1}^{N-1}\sum_{k=j+1}^{N}\delta_{jk}\right)\sum_{j=1}^{N-1}\sum_{k=j+1}^{N}\frac{\delta_{jk} - d_{jk^2}}{\delta_{jk}} \tag{15}$$

where, δ_{jk} = distance between two points in the new d-dimensional space, d_{jk} = distance between two points in the original d-dimensional space.

- *Backward Feature Elimination* [30] The meta-node employs the Backward Feature Elimination Technique to reduce dimensionality. A loop for Backward Feature Elimination is established within this meta-node, wherein a classification algorithm is trained and applied. Although Naive Bayes is the default algorithm, Swift can use another algorithm for classification if required. The Elimination End node finds and displays the error rate concerning the employed classifier. It sends control back to the Backward Feature Elimination Start which organizes the features provided to the classifier. When the process is done, the list of selected input features and the model of classification errors becomes available from the Backward Feature Elimination End node's output port. The Backward Feature Elimination filter allows counting how many features are chosen and what their corresponding error rates are after each iteration. A drawback of this approach is its limitation in terms of the maximum number of iterations, particularly for datasets with very high dimensions, which can result in extensive computation times [29].
- *Forward Feature Construction (FFC)* [30] Just as in Backward Feature Elimination, the forward feature construction loop proceeds by steadily including more input features in the creation of several classifiers. At first, only one feature is implemented

and the loop adds new features in the next iterations. The crucial features are chosen through a separate loop that delegates this task. The specific number of features is determined as a fixed parameter set within the Delegating Loop End node. The limitation of the Forward Feature Construction technique is analogous to that of the Backward Feature Elimination, primarily concerning the extensive computational time required for datasets with high dimensions.

2.1.4 Facial expression classification (FEC)

Classification happens as the last part of Facial Expression Recognition (FER). Among the methods used for this task are SVM, Generalized Discriminant Analysis (GDA), MLP-neural networks, k-NN, Random Forest, Hidden Markov Model and several others. They are used to label human emotions, picked out by the researcher depending on their study needs.

- *Support Vector Machine (SVM)* This is a recognized ML algorithm researchers frequently employ in various domains such as face recognition, FER systems, etc. It operates within the framework of supervised learning, extracting and categorizing patterns. The process involves mapping features into a high-dimensional feature space, which facilitates feature separation among different classes by optimising a hyperplane [1, 8]. The latest version of the MATLAB toolbox introduces a new library, called LibSVM[68] which is designed for polynomial kernels and radial bias functions.
- *Hidden Markov Model (HMM)* This approach permits the model of the facial action unit in order to aid in the recognition of facial expressions [29, 31]. Nevertheless, this model proves beneficial for voice recognition. Cao and Tong [32] propose a novel method using EHMM and LBP. The local characteristics are extracted using LBP, and the emotions are accurately classified using EHMM.
- *Random Forest (F_i)* Proposed by Breiman [69], this algorithm employs decision trees for data classification. Based on a greedy scheme, the algorithm develops N decision trees using the training subset and them merges the trees together to build a classifier that uses majority vote, stated in Eq. 16.

$$\mathrm{F}\left(\mathrm{x_i}\right) = \frac{1}{\mathrm{N}} \sum_{\mathrm{n}-1}^{\mathrm{N}} \mathrm{p_n}\left(\mathrm{x_i}\right) \tag{16}$$

Where $\mathrm{x_i}$ is the class $\mathrm{p_n}\left(\mathrm{x_i}\right)$ is the probabilities for each class.

- *Neural-Network Methods* Non-linear issues are the ideal fit for this neural network. Gueorguieva described a multilayer perceptron neural network for facial expression detection [33]. They used several networks to look for patterns in the data that represent emotions and concluded that the feed-forward neural network is the best at capturing both the S-shaped and radial bias functions [7, 34]. Backpropagation, Self-organizing map (SOM) and the Multi-Layer Perceptron (MLP) [7] neural network are available in the classification toolbox for neural networks provided in the toolbox.

In addition to the widely used classifiers such as SVM and Neural Networks, a variety of other supervised learning algorithms have been extensively explored for FER due to their

simplicity, interpretability, and effectiveness in handling different types of data. Logistic Regression is a fundamental statistical method that models the probability of discrete emotion classes using a logistic function, making it suitable for binary or multi-class classification problems where linear decision boundaries are acceptable. Decision Trees, on the other hand, provide a tree-like structure for decision-making by recursively splitting the data based on feature values, making them easy to visualize and interpret. However, they can be prone to overfitting, which is often addressed using ensemble methods like AdaBoost (Adaptive Boosting), which combines multiple weak classifiers to form a strong predictive model. AdaBoost adjusts the weights of misclassified samples during training to improve accuracy over successive iterations.

K-Nearest Neighbourhood (K-NN) is a non-parametric method that classifies new samples based on the majority class of the 'k' most similar training samples, usually determined by a distance metric such as Euclidean or Manhattan distance. K-NN is particularly effective for small and clean datasets, although its performance may degrade with high-dimensional or noisy data. Other ensemble techniques, such as Random Forests, which aggregate the predictions of multiple decision trees, and Gradient Boosting Machines (GBM), which optimize a loss function through iterative model updates, have also shown promising results in FER tasks. These methods are advantageous due to their robustness to overfitting and ability to handle both categorical and numerical features.

Furthermore, Naïve Bayes classifiers, which assume feature independence, and Extreme Gradient Boosting (XGBoost), a more efficient and regularized version of GBM, are increasingly being adopted in FER applications where speed and scalability are critical. The selection of an appropriate supervised learning algorithm depends on various factors, including the size and complexity of the dataset, the nature of the features extracted (geometric, appearance-based, or hybrid), and the computational resources available. By combining effective preprocessing and feature selection techniques, these algorithms can significantly enhance the accuracy and efficiency of FER systems across diverse real-world scenarios.

2.1.5 Emerging machine learning trends in facial expression-based user experience analysis

Recently, some studies [55, 65, 91, 92, 95, 101, 102, 109] implemented expression recognition with ML to determine user satisfaction towards a product or service. The proposed framework consists of three main components. The first step is collecting user satisfaction data based on gender, age, and facial expression information. This information is then classified using ML techniques. The last step is evaluating the model, which confirms that the model can achieve set instructions. Based on gender, age, and facial expressions, the classification model can predict user satisfaction [98]. This study concentrates on two important aspects. First, the design proposes a new geometrically based feature extraction technique that determines six distances to facial features that best express emotion. Second, the technique also implements a decision tree with the JAFFE dataset [99]. The main goal of this study is to build an automated system for facial expression recognition that can discern seven elementary emotions, including neutral ones, using a set of six distance measures, such as Euclidean or Manhattan distance. The system reached recognition rates of 89.20% for the JAFFE database and 90.61% for the COHEN database [100]. For this work, facial recognition is performed through PCA face recognition with maximum KNN and feature extraction with KNN. The dataset is

divided into training and test sets comprised of images of 10 female subjects captured through Japanese female facial expressions (JAFFE) with seven emotions: happy, sad, surprise, anger, disgust, fear, and neutral. The dataset included 210 images—140 for training and 70 for testing. The preprocessing steps included scaling, normalization, and converting the images into grayscale. The emotion features are C neutral, happy, sad, surprised, angry, disgusted, and afraid, while KNN is employed to classify them. The approach is implemented under MATLAB scripts and yielded an accuracy of 85.71% and 60 correct classifications out of 70 [101]. The study employed five-fold cross-validation and one-way ANOVA feature validation with $p < 0.01$ significance for final feature selection. These features constructed after cross-validation are used to train KNN and decision tree DT classifiers to recognize six emotions from facial expressions. KNN received an average accuracy of 98.03% by using four distance metrics. In contrast, DT received 97.21%. The development of different real-time systems electromagnetically is made possible by how well the optical flow algorithm can detect facial emotions [102]. The experimental analysis showed that the best SVM classifier is achieved with a recognition rate of 82.97%, followed by KNN with 79.99% and MLP with 82.97% [103]. PCA is utilized to reduce dimensional flexibility and extract hair-like features determined by eigenvectors. A basic kernel function unit and the SVM classifying unit are utilized to identify the primary emotions of happiness, sadness, and fear [104]. Combinations of four different algorithms are examined: SVM, Naive Bayes, Random Forest, and KNN: the results had various accuracy percentages; SVM earned 36\%, Logistic Regression got 64.2%, Naive Bayes counted 38.1%, and Random Forest 51.7% [105]. The proposed FER model requires three principal components: grid search optimization, data preparation, and classification. Linear Discriminant Analysis (LDA) placed the primary sentiments into eight categories: neutral, happy, sad, surprised, fear, disgust, anger, and contempt. Improved SVM attained an accuracy of 99% and F1 score of 98% [106]. A new technique is suggested to assess emotions that involve geometric features of the vectorized landmarks together with the curves around the landmarks. These features are from the regions of the human face that exhibit muscle movement.

SVM is used to identify features and parameters. Active learning and SVM are applied to classify facial action units [107]. The dataset comprised 11 emotions: normal, happy, sad, sleepy, wink, left light, surprised, no glasses, right light, glasses, and centre light. Metrics included precision, recall, accuracy, F1, and G-mean. The approach using the BGA-RF method attained an accuracy of 96.03, higher than all other methods [108]. While most systems perform well with controlled-environment datasets, they struggle with more challenging datasets due to increased variability and incomplete faces. A Histogram of Oriented Gradients (HOG) descriptor-based method is proposed, involving preprocessing and training an RF classifier. The JAFFE dataset demonstrated the method's effectiveness [109]. Performance evaluations based on confusion matrices and classification accuracy confirmed that combining SVM, PCA, and LDA features effectively identifies and classifies facial expressions [110].

Figure 12 presents an organized approach for constructing ML-based FER systems. The system begins its operations with Preprocessing steps that normalize raw images through resizing, filtering, and normalization techniques. The subsequent stage of Feature Extraction selects significant image attributes from pictures using PCA in combination with HOG and Haar-like feature extracting methods. Training Algorithms receive

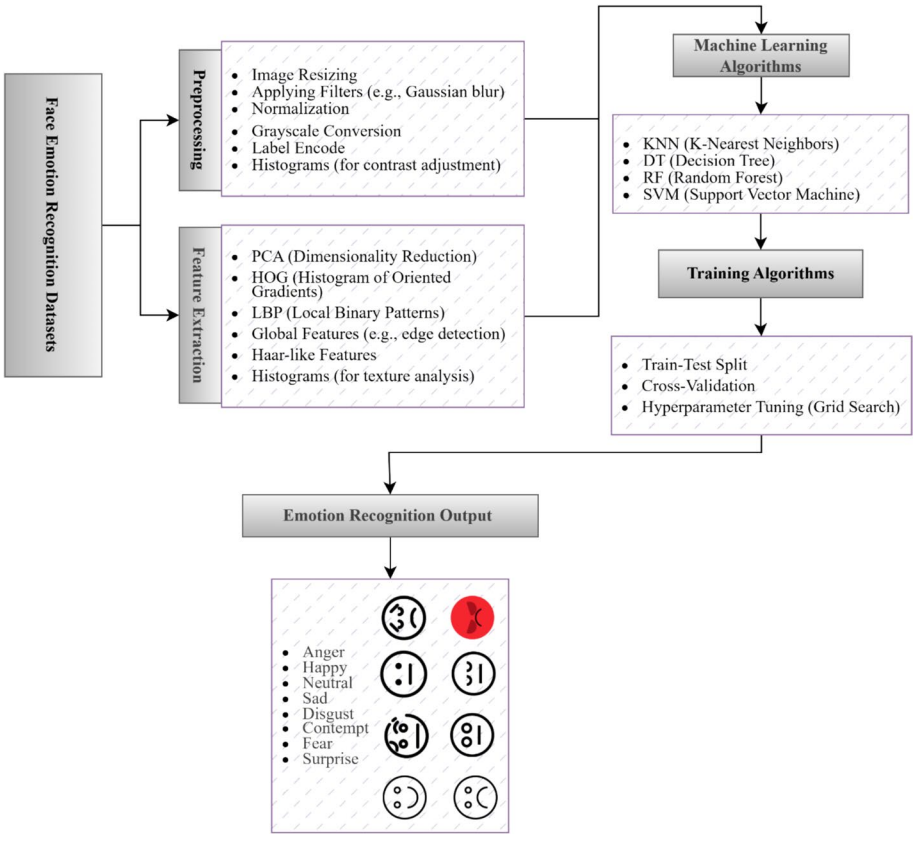


Fig. 12 Modular machine learning pipeline for facial emotion recognition

these features for processing ML models, including KNN, SVM and Random Forest, which are trained and optimized. As a final stage, the system generates an Emotion Recognition Output that identifies feelings of happiness, anger, sadness and surprise. The sequence of steps showcases the necessary data preparation process before emotion detection while explaining the importance of ML techniques for proper facial emotion analysis.

Table 3 presents the ML approaches to study FER systems. The studies describe several datasets, including CK + , JAFFE, Yale Faces, AffectNet, and others, to identify emotion variations from two to eleven. The techniques used are as varied as the emotions identified; for example, SVM, KNN, and RF, in addition to Logistic Regression, are the most common and are employed with a wide range of feature extraction methods. These methods include geometric and global features, PCA, LDA, LBP, Gabor filters, Haar-like features, HOG, and Haar Wavelet Transform. Some studies concentrated on direct classification, omitting intentional feature extraction, whereas others pursued a middle ground using several feature extraction techniques. Table 4 discusses the different studies, including key points such as specific feature extraction techniques and the success of optimization. Some studies have a narrow focus and aim almost exclusively at binary emotion classification.

2.1.6 Comprehensive review of machine learning approaches for facial expression recognition

Several studies have explored the application of ML approaches in the realm of facial expression recognition. These investigations focus on leveraging various ML techniques

Table 3 A review of machine learning methods for facial emotion recognition systems

Reference	Datasets Used	Methods	Feature Extraction Techniques	No. of Emotions Detected	Remarks/Key Findings	Issues and Limitations
[98]	Private Dataset	SVM	None	7	Focused on direct classification without explicit feature extraction	Lacks generalization due to no feature extraction and use of a private dataset
[99]	CK+, JAFFE	CART	Geometric and Global Features	7	Utilized decision trees with geometric and global features for emotion recognition	Decision trees are prone to overfitting and struggle with complex, high-dimensional data
[100]	JAFFE	KNN	PCA	7	Applied PCA for dimensionality reduction and KNN for classification	Sensitive to noise and suffers from scalability issues with large datasets
[101]	Private Dataset	KNN	Haar-like Features	6	Used Haar-like features for emotion detection with KNN	Performance may degrade with uncontrolled environments; private dataset limits reproducibility
[102]	CK+	SVM	LBP, Gabor Filter, PCA	7	Combined multiple feature extraction techniques for improved accuracy	High computational load and not easily applicable to real-time systems
[103]	CK+	SVM	PCA	7	Achieved high recognition rates using PCA and SVM	Limited robustness in natural, unconstrained settings with varying facial poses
[104]	FER 13	Logistic Regression	None	5	Direct classification without explicit feature extraction	Overly simplistic model; unable to capture complex, nonlinear facial patterns
[111]	AffectNet	SVM	Grid Search Optimization	8	Optimized SVM parameters using grid search for better performance	Computationally expensive optimization; limited adaptability to unseen variations
[106]	CK+, MUG	SVM	Geometric Features	8	Leveraged geometric features for emotion recognition	Highly sensitive to facial pose and illumination changes
[107]	CK+	SVM	PCA and LBP	7	Combined PCA and LBP for feature extraction	Performance drops in the presence of occlusions or dynamic expressions
[108]	Yale Faces	Random Forest	Histograms of Oriented Gradients	11	Achieved high accuracy using HOG features and Random Forest	Dataset has limited diversity; may not generalize to real-world conditions

Table 3 (continued)

Reference	Datasets Used	Methods	Feature Extraction Techniques	No. of Emotions Detected	Remarks/Key Findings	Issues and Limitations
[109]	JAFFE	Random Forest	Histograms of Oriented Gradients	7	Demonstrated effectiveness of HOG features in emotion recognition	Narrow demographic scope and controlled conditions reduce real-world applicability
[110]	FECR	SVM	LDA, PCA	6	Used LDA and PCA for dimensionality reduction and SVM for classification	Risk of losing critical information during dimensionality reduction
[112]	CK+	KNN	Histograms of Oriented Gradients	7	Applied HOG features with KNN for emotion detection	KNN is computationally expensive at inference and sensitive to irrelevant features
[113]	SPOS, BBC, MMI	SVM	Histograms of Oriented Gradients, LBP	2	Focused on binary emotion classification using LBP and HOG	Extremely limited emotion categories; lacks depth and general usefulness
[114]	CK+	SVM	Haar Wavelet Transformation and Principal Component Analysis	7	Combined Haar wavelet transform and PCA for feature extraction	Static feature extraction limits adaptability to spontaneous or subtle emotions

to enhance the accuracy and efficiency of facial expression recognition systems. The related work in this domain encompasses diverse methodologies, including feature extraction, classification algorithms, and model architectures. Researchers have extensively explored using DL models, such as CNNs and recurrent neural networks (RNNs), to learn discriminative features from facial images automatically. Additionally, studies have delved into the effectiveness of ensemble methods, SVMs, and other traditional ML algorithms for facial expression recognition tasks. This body of research contributes valuable insights into the ongoing efforts to advance the field and address the complexities associated with accurately detecting and interpreting facial expressions using ML techniques.

Alpesh Kumar [83] suggested PCA and distance classifier techniques to recognize facial expressions. PCA is used for facial expression detection for both training and testing image sets and generates the feature vectors. Furthermore, the result is obtained by applying Euclidean distance metrics. The Feedtum Facial expression database is used for experiment purposes. Seven facial expressions include anger, disgust, fear, happy, neutral, sad, and surprise. The result is defined with the help of the success rate and failure rate.

Olakean Agbolade [84] proposed the FER technique for the 3D dataset. The author suggested using multi-point warping for 3D facial landmarks and building a template mesh as a reference object. Procrustes ANOVA is used to evaluate the localization error. Feature selection is done using the PCA technique, and emotions are classified using the LDA method. Bosphorus and Stirling's dataset is used for experiment purposes. Six

Table 4 A comprehensive analysis of conventional FER

Authors & year	Emotions analyzed	Proposed methodology	Classifiers	Database	Training, validation and testing dataset	Result
Pawel[3]	Neutral, joy, surprise, anger, sadness, fear, disgust	Action units are computed through the Kinect device and employed as a feature	MLP	The Kinect device is utilized for 3D face modeling in conjunction with the KDEF dataset	For the 3-NN and MLP classifier, the training dataset is 70%, testing is 15%, and validation is 15%	The MLP classifier achieved a classification accuracy of 96% and 73% with a randomly divided dataset
Wenfei [35]	Happy, sad, surprise, disgusted, angry, scared and neutral	The Gabor filter is used to extract features, Gabor decompositions are encoded using radial grids, and PCA is used to reduce dimensionality	Modified kNN	CK and JAFE	Training data is JAFFE, and testing is done for the CK dataset	This approach outperforms the CK database, with an accuracy of 91.51%
D. Yang [15]	Sadness, happiness, surprise, fear, anger and disgust	Face detection is accomplished using the Haar cascade method while filtering and edge detection are done using Sobel edge detection	Neural Network	JAFFE	20 images have been taken for training	Accuracy and processing time is estimated
Benitez [36]	23 universal and compound emotions	The first step is to find the Euclidean distance between each landmark in the face area. Then, the angle that separates these landmarks is calculated, and Gabor filters are applied to the landmark points' centers	Kernel subclass discriminant	DISFA, CK+, and CE	Training dataset is 950 k AU annotated images, 88 k images for test data	The F1 score, accuracy, and mean error in AU intensity are calculated using the suggested technique
Kulkarni [15]	Happy, angry, Fearful, sad, disgusted, neutral	WLS technique is used for image pre-processing. Gabor and Log Gabor filters are used for feature evaluation. PCA is used for dimensionality reduction	SVM classifier	FACES	The training dataset is 80%, and for testing, 20% image dataset has been taken	An SVM classifier with the Log Gabor filter achieves an accuracy of 87%
Dhavalikar [62]	Happy, Sad, Anger, Fear, Disgust and surprise	The YCbCr model is employed for face detection, and features are extracted through the Active Appearance Model (AAM) method. The identification of human expression is then carried out using Euclidean distance	Artificial Neuro-Fuzzy Inference System	Real-time database	Not defined	The recognition rate approaches 90%-95% when a large number of training samples are utilized

Table 4 (continued)

Authors & year	Emotions analyzed	Proposed methodology	Classifiers	Database	Training, validation and testing dataset	Result
Mr. Mayur [64]	Angry, surprise, disgust, sad, smile, worry	Face recognition is based on skin color, and feature extraction involves scaling and distance measurements of individual facial features. After that, the curve's formation is assessed	Fuzzy Patterns	Real-time database	Not defined	The accuracy of expression is determined across various images in between 80–90%
Jianqiang [32]	Happy, sad, disgust, surprise, fear, anger and neutral	LBP is used for feature extraction	EHMM	JAFFE	Not defined	Offer 82%-92% accuracy for different parameters

types of emotions are used in this work. The accuracy for the Striling and Bosphorus datasets is 99.58% and 99.32%.

Tarik Boukhalfi [85] presented a novel approach to facial expression recognition. This work highlights the relationship between physical activity and facial expression. Subsequently, the model is trained using regression trees and SVM models that predict virtual characters' facial emotions from their 3D motion. However, this technique has some limitations because it addresses only one aspect of facial animation, i.e., physical activity.

The paper [86] has done the work with various methods and combined with ML techniques to obtain the results. Firstly, it detects the face from a given image using Viola Jonas, then extracts lips and mouth features through the Gabor filter, DWT, and HOG. Lastly, the SVM and Fisherface classifiers are applied and refined using Grid search to classify emotions. The CK + dataset is used, and six facial expressions are considered for evaluation. After computing the results with these methods, it is noticed that HOG and SVM produce better classifications. From this method, 85% accuracy is obtained.

Table 4 presents a comprehensive survey of conventional FER methods, highlighting diverse feature extraction techniques such as Gabor filters, LBP, and Euclidean distance calculations. Various classifiers, including SVM, MLP, and neural networks, have analysed emotions across datasets like JAFFE, CK + , and DISFA. The results indicate that while traditional methods achieve competitive accuracy, DL-based approaches often provide superior performance through advanced feature learning and classification.

2.2 Deep learning based FER tactics

Deep learning-based facial expression recognition (FER) leverages neural networks to automatically learn facial features for emotion classification. It processes raw facial images to identify subtle expression variations. Large-scale annotated datasets enhance model accuracy and generalization. Real-time FER systems integrate DL for applications in healthcare, surveillance, and human–computer interaction. Continuous advancements improve robustness against occlusions, lighting changes, and pose variations.

DL has become popular in various computer vision domains in recent decades. These topics cover a wide range of tasks, such as feature extraction, classification, object identification [37], face recognition [38], scene understanding, and facial expression

recognition [13]. This section briefly overviews CNN and RNN architectures, which have seen widespread application among academics and research in this field. CNNs excel in image modelling, but RNNs excel at sequence modelling. The structure of the human brain inspires these DL networks and is useful in various computer vision tasks.

2.2.1 Convolutional neural network (CNN)

In 1998, LeCun [39] introduced the CNN technique. Initially, an image or signal is convolution with filters within the convolutional layers. This technique produces a feature map, each element identifying a distinct feature across various input locations [12]. Zeiler [40] describes a method for visualising a single unit in a CNN's feature set (or map) using a de-convolutional network trained on the same dataset. When this network is run through the convolutional layer, the input pixels responsible for activating a given neuron can be visualised. The feedforward step of this method simulates a filter attempting to maximise its output. Numerous networks within the CNN architecture can accurately identify and classify objects, humans, and scenes with minimal processing, as demonstrated in the ImageNet challenge. These networks differ based on their applications and often compete against each other. The upcoming Table 5 provides an overview of the various types of CNN networks.

ReXNeXt is a considerable improvement over ReXNet, emphasising increased cardinality to improve network performance rather than simply increasing network width or depth. Experimentation has shown that increasing cardinality considerably reduces the total error rate, resulting in an optimised network across multiple applications.

As shown in Fig. 13, the CNN architecture comprises multi-stage processing of input data to extract intricate patterns or deep knowledge. CNNs are modelled after real neuron networks, in which neurons are represented as weights organised in three-dimensional space and defined by height, width, and depth. DenseNets is another modern network and leads to better performance in minimum complexity compared to ResNet models.

There are four processing layers in CNN: (i) Convolutional layer, (ii) Pooling layer, iii) Rectified Linear unit layer, and (iv) Fully connected layer [41], as discussed as follows.

2.2.1.1 Convolution layer A trainable filter performs spatial convolution on a set of N input neurons. This procedure involves computing the dot product of the filter's weights

Table 5 Different types of CNN Network

T Types of CNN Network	Year	Developed By	Number of Layers	To Top-5 Error rate	Number of parameters	Types of Applications
LeNet	1998	Yann Lecunn et al	5	–	60 thousand	Identify handwritten digits
AlexNet	2012	Alex Krizhevsky et al	7	15.3	60 million	ImageNet classification
ZNet	2013	Matthew Zeiler	8	14.8	–	Image classification
GoogleNet/ Inception	2014	Google	19	6.67	4 million	Deeper classification
VGG Net	2014	Simonyn Zisserman	22	7.3	138 million	Large-scale image recognition
ResNet	2015	Kaiming He	152	3.6	25 million	Large-scale image recognition

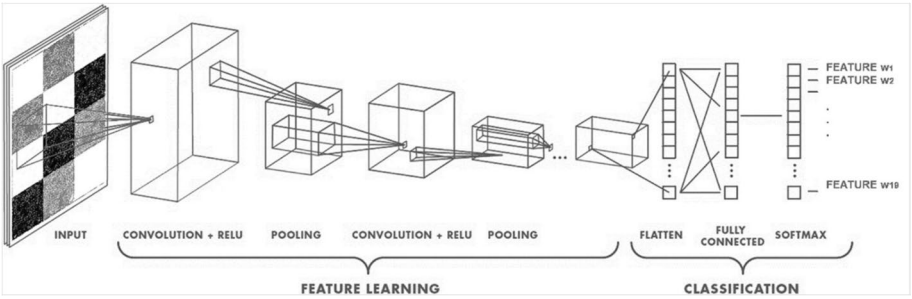


Fig. 13 Architecture of CNN model

and the input neurons, resulting in feature maps that are then transmitted to the next layer. The major goal of this layer is to fine-tune the filter weights to identify the most important data features. This layer is associated with several essential words, including Kernel Size, Stride, Depth, and Padding, each of which serves a distinct purpose.

2.2.1.2 Pooling layer The goal of this layer is to minimise the dimensions of input maps while achieving rotational (low) and translational invariance, guaranteeing that deviations mirror the input. The pooling procedure is used in CNNs to calculate the average or maximum value within a spatial area. Both of these approaches are described in Eqs. (17) and (18) [1].

$$\widehat{\mathrm{f_{avg}}}\,(\mathrm{m}) = \frac{1}{\alpha \times \beta} \times \sum \mathrm{f}\,(\mathrm{u} : \mathrm{u} + \alpha, \mathrm{v} : \mathrm{v} + \beta, \mathrm{k}) \tag{17}$$

$$\widehat{\mathrm{f_{max}}}\,(\mathrm{m}) = \mathrm{MAX}\,(\mathrm{f}\,(\mathrm{u} : \mathrm{u} + \alpha, \mathrm{v} : \mathrm{v} + \beta, \mathrm{m})) \tag{18}$$

where the dimensions of the spatial window are α and β, u and v are the locations of input slice m. f(u,v,m) is to obtain the local patch from the precedence layer, the Pooled pixel at slice m presented by $\widehat{f_{avg}}\,(m)\; and \widehat{f_{max}}(m)$.

2.2.1.3 Rectified linear unit layer (ReLU) The activation function is used within this layer to introduce nonlinearity to the neurons. The ReLU layer eliminates negative neurons by using the max function on the neurons and a threshold of zero, as shown in Eq. (19).

$$\mathrm{F}\,(\mathrm{x}) \;= \max\;(0,\;\mathrm{x}) \tag{19}$$

An enhanced version of ReLU is Leaky ReLU, which has introduced by He [42]. This new version incorporates a small negative slope, preventing all negative neurons from vanishing (instead of always being set to zero). It also offers the flexibility of being parametrized, allowing for adjustments according to the specific requirements for achieving improved outcomes.

2.2.1.4 Fully connected layer After undergoing various operations within the max pooling and convolutional layers, the neural network engages in high-level reasoning through the fully connected layer. This layer involves all connections with the activation function from the preceding layer, ultimately condensing into a 1×1 spatial size. The depth of the data gets consolidated throughout this process. The corresponding equation is provided below:

$$x_j = f\left(\sum_{i=1}^{m} z_i.\ w_{i,j} + b_j\right) \tag{20}$$

where,

x_j = calculating the function f(z) of given input z_i from the previous layer and produced output neuron x_j.

j = individual weight of a fully connected layer.

b_j = bias of the connected layer.

Dropout aids in mitigating overfitting by being implemented following the final convolutional layer and between the fully connected layers, including certain probability-based values. The concept of dropout probability dictates that the output of each neuron is assigned a value based on a given probability [43].

2.2.1.5 Classification layer The assessment of a fully connected layer's performance relies on its anticipated probabilities, which are achieved through the utilization of a classification layer. The Softmax function [12] is employed in this layer to classify multiple categories. Equation (21) illustrates the Softmax function's formula, wherein it takes the CNN's outcomes as a vector of scores (denoted as "z") and maps them to a range between 0 and 1. This ensures that these values' sum equals 1 [12].

$$S_{\mathbf{n}}(z) = \frac{e^{z_n}}{\sum_{i=1}^{N} e^{z_i}} \tag{21}$$

Cross entropy is then calculated as shown in Eq. (22)

$$L = -S_{y_i} + \log\left(\sum_{n=1}^{N} e^{S_n}\right) \tag{22}$$

2.2.2 Recurrent neural network (RNN) or long short term memory (LSTM)

A novel hybrid technique is proposed, which combines CNN for capturing spatial features inside individual frames and LSTM for capturing temporal features over successive frames. This fusion is required since the CNN technique alone is insufficient for addressing temporal changes in facial components. The primary purpose of using LSTM is to tackle long-term dependency issues by leveraging its short-term memory capabilities. LSTM is a specialised form of Recurrent Neural Network (RNN) designed to aid in learning extended dependencies. While RNN networks use a chain-like structure with repeated modules [72], LSTM also uses a chain-like structure. The structure of LSTM is illustrated in Fig. 14. Here, An LSTM modifies the information stored in the cell state by either increasing or decreasing it. The forget gate layer decides which information to keep or discard, the input gate layer updates the cell state with selected values, and the output gate layer produces results based on the updated cell state.

RNN or an LSTM has two advantages over a standalone-based technique for modelling consecutive pictures [44].

- The LSTM offers a simple method for fine-tuning end-to-end learning as compared to the other CNN models.
- Suitable for both fixed and variable-length inputs and outputs, LSTM compatible.

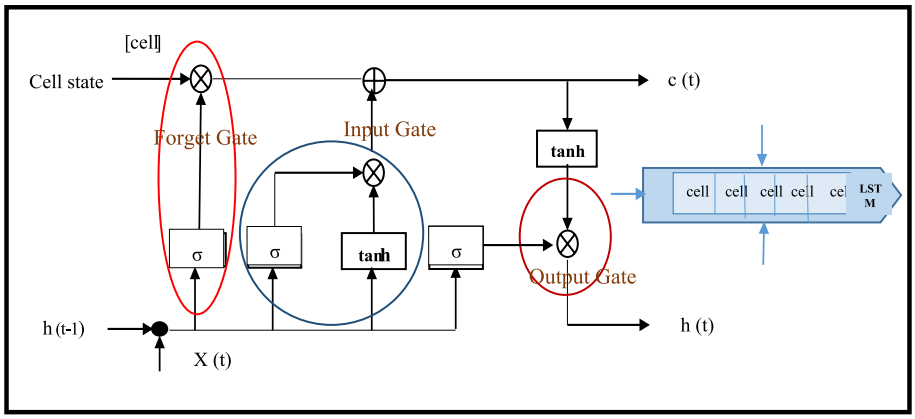


Fig. 14 LSTM's architecture

Apart from RNN and LSTM, other recurrent units like Gated Recurrent Units (GRUs) and their variants have shown considerable promise in Human Activity Recognition (HAR) tasks due to their computational efficiency and ability to model temporal dependencies. GRUs, a simplified version of LSTMs, combine the forget and input gates into a single update gate and merge the cell and hidden states, making them faster to train while maintaining comparable performance. Their lightweight architecture is particularly beneficial in real-time and resource-constrained HAR scenarios, such as wearable and mobile devices. Moreover, Bidirectional GRUs (Bi-GRUs) and stacked GRU architectures have been employed to capture both past and future context in sequential data, improving the recognition of complex activity transitions. Additionally, hybrid models integrating CNNs with GRUs are widely used to extract spatial features through convolutional layers and model temporal dynamics using GRUs, offering a robust framework for spatiotemporal HAR tasks. These models have demonstrated high accuracy on benchmark datasets like PAMAP2, UCI-HAR, and Opportunity, making GRU-based architectures a compelling alternative to traditional RNNs and LSTMs for efficient and effective human activity recognition.

2.2.3 Emerging deep learning trends in facial expression-based user experience analysis

The last twenty years should be marked with progress for artificial intelligence and computer vision researchers with the development of Fer or facial expression recognition systems. FER is crucial for the interaction between humans and machines since faces are not the only reliable cues for expression recognition; the non-facial features and background of the images are equally important. The welfare of the mobile network population has shifted towards the need for highly responsive and effective systems. Unfortunately, most of the existing FER networks fail to do so because of the real-time constraints combined with a high level of computational complexity and numerous parameters. In this regard, the researchers developed a new method that implements supervised transfer learning, island loss, and joint-supervision. The method, which uses supervised transfer MobileNetV2, an efficient and precise CNN model, is claimed to yield real-time results and is reported to outperform many other competing techniques on CK+, JAFFE, and FER2013 databases. Another claim comes from a more specific model, A-MobileNet, which claims to achieve higher local feature acquisition by applying the attention module in MobileNetV1. Using the combination of centre loss and

softmax loss, the model can advance without raising the number of the model's components. This has been shown on recognition tests for the RAF-DB and FERPlus datasets. These models, while efficient in certain parts of the world, are not easy to use globally. It has been suggested that several countries' datasets be pooled into one to create a more accurate model of FER. As a proof of concept, the researchers trained a convolutional neural network (CNN) model on JAFFE and CK databases that are designed to be complex, resource-efficient, and quick to train [117]. The advances in VGG16 have completely transformed the possibilities in facial expression recognition (FER), which utilizes DL techniques. Specifically, replacing fully connected layers with convolutional ones and utilizing adaptive pooling. These changes resulted in a boost of 7% in accuracy on the RAF-DB and SFEW datasets [118]. Moreover, DL models have also been developed for real-time engagement detection in online classes using Inception-V3, VGG19, and ResNet-50. ResNet-50 outperformed all these models with an accuracy of 92.3% on CK + , RAF-DB and FER-2013 datasets [119].

Real-time facial expression recognition systems, based on Haar cascade detection and CNNs, have been created to recognize emotions such as happiness, anger, and surprise [120]. Some newer models like ResNet-50 and the residual masking networks have excelled in FER with accuracy on par with the best in the world in datasets FER2013 and VEMO [121, 122]. Evaluations on JAFFE, CK + , and FER2013 for accuracy, precision, recall, and F1-score show these methods are effective and more advanced than other known methods [123]. The FERC model identified five facial expressions and achieved an accuracy of up to 96% by adopting an expressional vector EV [124]. Further, DenseNet-161 achieved 96.51% and 99.52% accuracy on KDEF and JAFFE datasets, respectively, marking another milestone [125].

Attention-based convolutional networks have also been proposed, significantly outperforming previous models on datasets like FER-2013, CK + , and JAFFE [126]. DL frameworks, including DCNNs, have been developed to identify five emotional states, achieving 70% accuracy on FERC-2013 and 98.65% on JAFFE [127, 128]. These methodologies are increasingly significant in classification tasks, particularly in challenging scenarios like low-light conditions or masked facial images [129, 130]. FER has diverse applications, including security, healthcare, and safety assessments [131]. For instance, real-time emotion detection systems have been developed to identify autistic individuals [132]. Engagement detection systems using DL models have also been proposed for online education, evaluating learners' facial expressions to measure engagement levels [133]. The FERC model, utilizing CNNs, classifies facial expressions into five categories by analyzing key facial landmarks [134]. Innovative approaches, such as feature extraction with CNNs and hierarchical attention networks, have further improved FER accuracy and stability [135, 136]. Despite significant progress, FER remains complex due to the dynamic nature of facial expressions [137]. MobileNetV1-based methods have been proposed to enhance classification accuracy in challenging scenarios [138]. Additionally, hybrid architectures combining CNNs and vision transformers (ViT) with CoT modules have been developed to detect subtle facial variations in complex environments [139]. Other advancements comprise selective kernel networks [140] and Cross Centroid Ripple Pattern Method CRIP for FER using computer technology [141].

These innovations highlight the ongoing evolution of FER systems, driven by DL and feature extraction techniques, to achieve higher accuracy and real-time performance

across diverse applications. The figure (as shown in Fig. 15) depicts a well-organized process of recognizing facial emotions, which is initiated by preprocessing activities which include image resizing, augmentation (flipping, rotation, and distortion), normalization (grayscale, histogram equalization), and the addition of Gaussian noise. These are the steps that ready the data for training. The training stage uses hyperparameter optimization and is based on multiple architectures of deep neural networks (DNN), including CNN, MobileNet-V1/V2, DenseNet-169, ResNet-50, Inception-V3, and VGG-16/19. Models are built robustly, and at the last stage, the recognized emotions, which include anger, happiness, sadness, disgust, contempt, fear, and surprise, are delivered. This flow illustrates the combination of preprocessing techniques, advanced neural networks, and emotion classification for facial emotion recognition, which makes the system accurate and efficient.

Recent advances in DL-based Human Activity Recognition (HAR) [51, 85, 107, 168–171] have been significantly influenced by emerging technologies such as Vision Transformers (ViT), Explainable AI (XAI), lightweight model architectures, and vision-language models. ViTs have introduced a novel approach by leveraging self-attention mechanisms to capture long-range dependencies in activity data, improving recognition accuracy over traditional convolutional networks. Meanwhile, XAI methods have become crucial in HAR to interpret model decisions, providing transparency and trustworthiness, especially in sensitive applications like healthcare and security. Alongside, developing lightweight models addresses the challenge of deploying HAR systems on resource-constrained devices, balancing performance with efficiency. Furthermore, integrating vision-language models enriches HAR by combining visual inputs with semantic understanding, enabling more context-aware and robust activity recognition. Together,

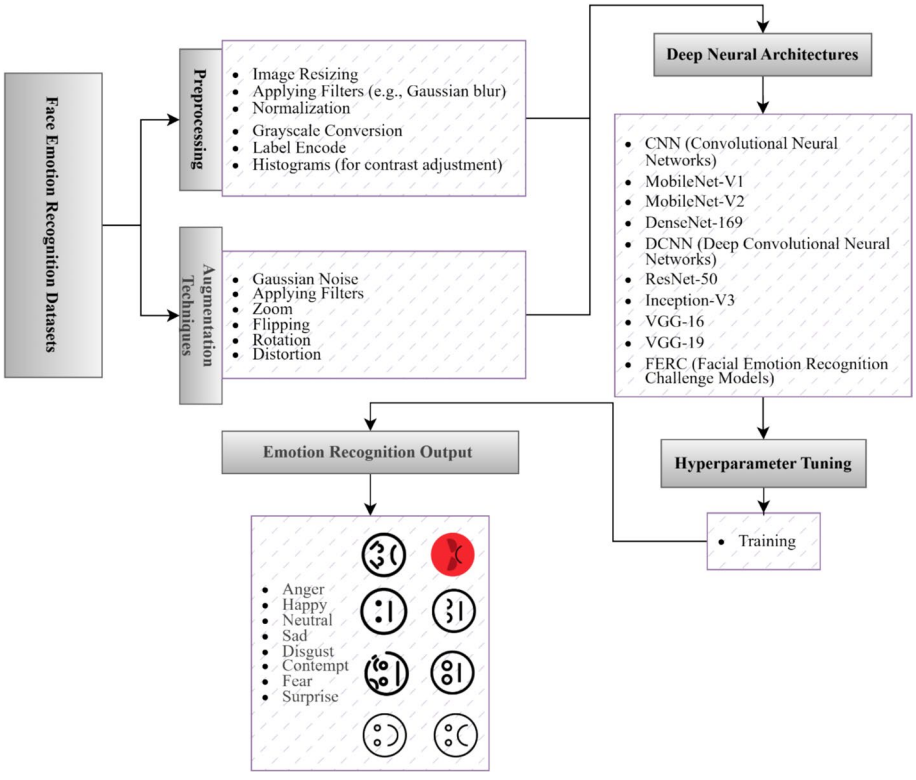


Fig. 15 Comprehensive workflow for facial emotion recognition using deep learning

these trends are reshaping the landscape of HAR, pushing the boundaries toward more accurate, interpretable, and deployable systems. The detailed discussion is as follows.

2.2.3.1 Vision transformer (ViT) in human activity recognition The Vision Transformer (ViT) [158–161, 163] and its variants have rapidly become powerful tools in Human Activity Recognition (HAR), especially for vision-based modalities. Unlike CNNs, which rely on local receptive fields, ViTs model long-range dependencies using self-attention mechanisms. In HAR tasks involving RGB or depth video inputs, the input sequence is decomposed into fixed-size, non-overlapping image patches (e.g., 16×16 pixels). These patches are linearly embedded into vectors (e.g., 768-dimensional in ViT-Base) and augmented with positional encodings to retain spatial structure. The [CLS] token is appended to the sequence and serves as the holistic representation for classification. The embedded patch tokens then pass through multiple Transformer encoder layers, each comprising Multi-Head Self-Attention (MHSA), Layer Normalization, and Feed-Forward Networks (FFN) with GELU activation. These layers are highly effective at modeling global spatial–temporal relationships across video frames, enabling ViT to capture subtle motion cues in complex human activities. Datasets like NTU RGB + D, Kinetics-Skeleton, and HMDB51 have shown substantial performance improvements when ViT replaces traditional CNN backbones.

Recent developments have introduced specialized ViT models tailored to spatio-temporal HAR, including TimeSformer, MotionFormer, and ViViT. For instance, TimeSformer extends ViT by decoupling spatial and temporal attention, allowing it to process longer video sequences efficiently. MotionFormer incorporates temporal difference encoding to focus on inter-frame motion, improving recognition of dynamic activities such as running or jumping. Meanwhile, ViViT (Video Vision Transformer) explores several configurations, such as factorized encoders for spatial and temporal attention, and has demonstrated strong performance in video-based HAR. These models also leverage learnable temporal embeddings to account for the sequence of frames, making them suitable for long-duration activity recognition. Advanced hybrid architectures, such as Conformer-HAR and CNN-ViT hybrids, combine the local pattern extraction capabilities of CNNs (e.g., ResNet, Inception) with the global modeling of transformers. These configurations help overcome the limitation of ViT's limited inductive bias, making them more data-efficient and better suited for real-world, low-data scenarios.

In sensor-based HAR, ViT models have been adapted by transforming 1D sensor signals (e.g., accelerometer or gyroscope data) into 2D image-like representations such as Gramian Angular Fields (GAF), Recurrence Plots, or Spectrograms, enabling direct application of vision models. These representations are tokenized into patches and processed through ViT or hybrid PatchTST-like models that integrate temporal embeddings to preserve time dependencies. Alternatively, SensorViT and similar architectures introduce axis-specific tokenization (e.g., treating x, y, and z-axis signals as separate channels or tokens) and cross-channel attention, enhancing interpretability and performance. In some cases, CNN backbones are used to extract short-term local patterns before feeding token sequences into a ViT encoder to model long-term dependencies. The final [CLS] token is passed through a fully connected classification head with softmax output. These architectures achieve competitive results on benchmark datasets like PAMAP2, UCI-HAR, WISDM, and Opportunity, outperforming RNN-based methods in recognizing

both fine- and coarse-grained activities. Moreover, ViT-based HAR systems are increasingly being integrated with explainability tools, such as attention heatmaps and token contribution maps, offering greater transparency in model decision-making.

2.2.3.2 Explainable AI in human activity recognition Explainable AI (XAI) in Human Activity Recognition (HAR) [161–163] plays a critical role in enhancing transparency, interpretability, and trust in models that are often deployed in sensitive and high-stakes environments such as healthcare monitoring, security surveillance, and human–robot interaction. HAR systems, particularly those based on DL architectures like CNNs, Recurrent Neural Networks (RNNs), Graph Convolutional Networks (GCNs), and Transformers, often function as black boxes obscuring their decisions' rationale. XAI methods address this opacity by offering insight into the internal reasoning of these models, identifying which input features, time steps, or spatial regions most influenced a prediction. This transparency helps users validate results, uncover biases, and debug erroneous predictions. For example, in fall detection systems for elderly care, explainable models can clarify whether a true fall occurred or if the model misinterpreted a sudden posture shift.

So far, Explainable AI techniques have been integrated into prominent HAR models to address the interpretability challenge. For instance, CNN-based models such as DeepConvLSTM and HARNet have been coupled with gradient-based XAI techniques like Saliency Maps, Integrated Gradients, and SmoothGrad to highlight influential time-series segments and sensor channels. Transformer-based models such as TimeSformer, HARFormer, and PoseFormer employ attention visualization and rollout techniques to interpret spatiotemporal attention dynamics. The attention dynamics reveal frames, patches, or joints that influence predictions. In vision-based HAR, I3D (Inflated 3D ConvNet) and SlowFast Networks have been combined with Grad-CAM (Gradient-weighted Class Activation Mapping) to highlight discriminative regions in video frames, allowing analysts to visualize what the model "sees" when recognizing activities. Furthermore, Graph Convolutional Networks (e.g., ST-GCN, 2 s-AGCN) used for skeleton-based HAR have adopted node-level attention heatmaps, enabling joint-level interpretability, particularly for gesture and action recognition tasks.

These explainable models have found practical applications across domains. In healthcare, explainable RNN or Transformer models provide clinicians with justifications for triggering alerts, improving confidence in real-time monitoring systems. In manufacturing and workplace safety, interpretable GCNs can show which joints contributed to recognizing dangerous movements, informing ergonomic policy improvements. Visual feedback from models like PoseFormer or ST-GCN in sports and rehabilitation uses attention overlays to analyze an athlete's form or movement inefficiencies. Despite these advances, several challenges remain, such as aligning model explanations with human reasoning, handling multimodal data fusion explainability, and creating standardized benchmarks for XAI in HAR. Future directions include integrating post-hoc explainability (e.g., SHAP, LIME) into end-to-end HAR pipelines, developing intrinsically interpretable architectures, and refining human-in-the-loop frameworks to incorporate expert feedback into model explanations.

2.2.3.3 Lightweight and On-Device HAR models The push toward lightweight and on-device HAR models [173–176] is motivated by the need to deploy real-time activ-

ity recognition systems in portable, wearable, or embedded platforms where computational resources, power consumption, and memory footprint are highly constrained. Traditional DL models, such as large CNNs or Transformers, are typically too resource-intensive to run efficiently on such devices. To bridge this gap, models like MobileNetV2, ShuffleNet, and EfficientNet-lite have been adapted for HAR tasks. These architectures utilize design strategies like depthwise separable convolutions, inverted residual blocks, and squeeze-and-excitation modules. Such modules significantly help to reduce the number of parameters and floating point operations (FLOPs). In sensor-based HAR, these models have been retrained or fine-tuned using datasets such as WISDM, PAMAP2, and RealWorld HAR, achieving competitive accuracy while maintaining low inference latency (e.g., < 20 ms per sample on ARM Cortex-M processors).

Expanding on lightweight attention-based models, architectures like MobileViT, Lite Transformer, and TinyViT integrate global context modeling into compact networks. MobileViT, in particular, merges 2D convolutions with small self-attention modules that operate on local feature maps, allowing it to capture both spatial and temporal correlations in activity sequences. In the context of HAR, MobileViT has shown promise in both vision-based (e.g., video frame) and sensor-based (e.g., accelerometer signal) modalities. Its hybrid structure allows it to perform better than pure CNNs in recognizing complex activities that require understanding long-range temporal patterns. These patterns includes transitioning from walking to sitting or detecting anomalous behavior in surveillance systems. Moreover, these models can be accelerated further using compiler-level optimizations (e.g., TensorRT, TVM) and hardware-aware pruning methods that preserve performance on specific edge AI chips (e.g., Google Edge TPU, NVIDIA Jetson Nano).

Another emerging category includes custom-built HAR-optimized models such as LiteHAR, HARNet, MicroHAR, and TinyHARNet. Such models are designed using manual or automated techniques like Neural Architecture Search (NAS). These models often employ 1D convolutional layers to process time-series sensor data efficiently. The models incorporate modules like temporal attention, dilated convolutions, and residual shortcuts to enhance recognition of fine-grained activity patterns without introducing computational overhead. For instance, HARNet utilizes a multi-scale convolutional block and channel-wise attention to extract discriminative features from multi-sensor inputs. Other side, MicroHAR uses NAS to automatically discover the best lightweight structure for target devices, balancing memory, FLOPs, and latency constraints. These networks are typically benchmarked using edge metrics such as Energy Delay Product (EDP). They can run inference in real-time (e.g., < 10 mW power consumption) on devices like Arduino Nano 33 BLE Sense or Raspberry Pi Pico. Their development marks a shift toward device-aware model design, ensuring that HAR systems are accurate and deployable in real-world, mobile, and always-on scenarios.

2.2.3.4 Vision language models in human activity recognition Recent advances in Vision-Language Models (VLMs) [163, 165, 167, 168] have opened new avenues for Human Activity Recognition (HAR) by integrating visual perception with semantic understanding. Traditional HAR approaches often rely solely on CNNs for spatial feature extraction and recurrent neural networks (RNNs) or temporal convolutional networks (TCNs) for modeling temporal dependencies in video sequences. In contrast, VLMs like CLIP (Con-

trastive Language–Image Pretraining) and VideoCLIP incorporate joint image-text representation learning, enabling zero-shot recognition of human activities. These models are trained on large-scale image or video–text pairs using contrastive learning, aligning embeddings from both modalities in a shared latent space. As a result, they can match video frames to textual descriptions of actions (e.g., "a person doing yoga" or "a man riding a bike"), even if the model has not seen those actions during training.

The architecture of VLMs used in HAR typically involves a visual encoder (e.g., ResNet, Vision Transformer) and a text encoder (e.g., Transformer-based models like BERT or GPT variants). During inference, video frames or clips are processed using the visual encoder to generate feature embeddings, while candidate action labels are embedded using the text encoder. These embeddings are then compared using cosine similarity or a dot-product scoring function to determine the most semantically aligned action. Extensions of these models for HAR often integrate temporal context via 3D ConvNets temporal transformers to handle the motion dynamics across frames. Additionally, prompt engineering and textual template design are used to enhance the expressiveness of textual queries (e.g., "a person is [action]"), improving model performance in few-shot or zero-shot settings.

Applications of VLM-based HAR span surveillance systems, assistive robotics, healthcare monitoring, and multimedia content analysis. These models provide key benefits such as transferability to unseen activities, robustness to domain shifts, and interpretability through natural language explanations. Furthermore, some recent research incorporates fine-tuning techniques such as cross-modal attention fusion, adapter modules, and temporal prompt tuning to improve VLM performance on HAR-specific datasets like Kinetics, Something-Something, and Charades. Models like ActionCLIP and EGO-CLIP demonstrate how integrating spatiotemporal encoders and structured activity descriptions can significantly outperform traditional CNN + RNN baselines. As the field evolves, combining VLMs with knowledge graphs, scene graphs, and contextual metadata offers promising directions for achieving a deeper understanding and generalization in activity recognition.

2.2.4 Comprehensive review of deep learning approaches for facial expression recognition

Weiqing Wang [81] introduced a new framework tailored for assessing facial expressions in online platforms, a particularly relevant system amidst the COVID-19 context due to the widespread adoption of online schooling. This system utilizes cameras integrated into laptops and mobile devices to capture students' facial images, subsequently analyzing and categorizing their expressions into eight distinct emotional types. The images undergo preprocessing through IntraFace, facilitating facial feature tracking, attribute detection, and head pose estimation. Following the preprocessing steps, a DL model built upon the Convolutional Neural Network (CNN) architecture is applied to acquire a grasp of the features and classify human emotions. The results are then subjected to analysis. This proposed model underwent training using Jaffe, CK+, and FER2013 databases.

Ramachandram Vedanthan [80] introduced an innovative approach to facial expression recognition, emphasizing the importance of robust feature extraction through an optimized DBN-SMO technique for FER. The database undergoes pre-processing to eliminate illumination and lightning effects from the images. Subsequently, LPQ, WLD,

and LBP methods are employed for feature extraction. Incorporating the DCT technique further enhances feature extraction, improving recognition rates while minimizing computational costs. Additionally, PCA is utilized to reduce dimensionality. In the final stages, a Deep Belief Network (DBN) is integrated with the Spider Monkey Optimization (SMO) technique to classify facial emotions effectively. The experimentation employs KDEF, MMI, CK+, DISFA, and Oulu-CASIA datasets. Remarkably high classification accuracy rates are achieved, specifically 97.39%, 95.42%, 97.58%, 95.76%, and 92.38% for the seven classes of emotions, respectively.

Shekhar Singh [82] introduced a Facial Expression Recognition (FER) method built upon the foundation of a CNN model. The initial phase involves preprocessing to execute various face detection and alignment tasks. Face detection is accomplished using the Haar cascade classifier. Subsequently, facial features, including eyebrows, eyes, nose, upper and lower mouth regions, and jaws, are extracted. In the final stage, the CNN architecture comes into play for emotion classification. The FER2013 dataset is utilized for assessment. Notably, an accuracy of 61.7% is achieved without the utilization of preprocessing or feature extraction methods. With the application of an ensemble of CNN in a cutting-edge classification setup, the accuracy further increases to 75.2%.

Kahou et al. [45] proposed a hybrid CNN-RNN system and detailed a comprehensive framework for the Emotion Recognition in the Wild (Emotiw) challenge [46]. They demonstrated that an integrated CNN-RNN architecture for emotion analysis outperforms the previously employed CNN-based approach, which relied on temporal averaging for accumulation. This architecture is adept at conveying information across a sequence by means of a hidden layer representation.

Face sequences on the apex, offset, and onset expression condition are found by Kim et al. [10] instead of expression intensity. Using CNN, the spatial properties of the picture of the demonstrative expressive state are conveyed. Temporal aspects of spatial feature representation are completed in the second portion, whilst the first part is accomplished using an LSTM of face expression.

Hasani and Mahoor [47] introduced a 3D Inception—ResNet framework coupled with an LSTM model. This combined approach effectively captures both the spatial and temporal relationships within facial images across distinct frames within video sequences. Instead of focusing on specific facial regions, their network takes facial landmark points as input and emphasizes the key components of the face. To address temporal dependencies within image sequences during classification, the authors employed a recurrent network method proposed by Graves et al. [48]. The experimental outcomes showcased two types of LSTM: bi-directional and uni-directional LSTM. It is demonstrated that the bi-directional LSTM network outperforms the uni-directional LSTM network in terms of both results and performance.

Mao et al. [158] proposed AU-ViT, an AU-aware Vision Transformer that addresses label bias and domain shift in facial expression recognition (FER). By incorporating a dedicated AU branch, the model improves representation learning and robustness to occlusions. AU-ViT achieves superior or comparable results to state-of-the-art methods across multiple FER datasets. Hosseini et al. [159] analyze bias in facial expression recognition datasets and models, highlighting fairness issues across demographic groups. Their study reveals that model accuracy varies significantly by ethnicity and gender, with some groups experiencing up to a 15% lower accuracy. The authors emphasize the need

for more balanced datasets to improve fairness and reduce performance disparities. Based on the subgroup and dataset, reported accuracy ranges from 70 to 90%. Ren et al. [160] propose an improved YOLOv8 model enhanced with transformers and data augmentation for infant facial expression recognition. Their approach achieves high precision, with accuracy exceeding 92% on a custom infant dataset. The model demonstrates faster detection speed and better robustness compared to previous methods. This work advances DL applications in pediatric facial expression analysis. Cruz-Vázquez et al. [161] propose an advanced DL model for emotion recognition from EEG signals, leveraging quantum rotations and Fourier Neural Networks for enhanced feature extraction. Their convolutional neural network (CNN) architecture achieved a 95% accuracy, particularly excelling in identifying sad emotions. This approach demonstrates the efficacy of integrating quantum transformations with traditional signal-processing techniques in EEG-based emotion classification. The study highlights the potential of combining quantum computing methods with DL to improve the accuracy and robustness of emotion recognition systems.

Yang et al. [162] introduced a visible light human activity recognition framework leveraging a generative language model to bridge vision-language gaps in semantic understanding. Their approach integrates visible spectrum imagery with activity prompts using a text-conditioned generative model to enhance feature alignment. The method achieved 94.21% accuracy on VLP-HAR and outperformed conventional CNN-LSTM and Transformer-based baselines. The results validate the potential of generative VLMs in zero-shot and few-shot activity recognition under visible lighting conditions. Zhang et al. [163] proposed PeVL, a Pose-Enhanced Vision-Language Model that integrates human pose features into CLIP for fine-grained action recognition. The model employs a graph-based pose encoder and pose-conditioned prompt generator to improve visual-language alignment. PeVL achieved 90.3% accuracy on FineGym and outperformed prior methods on Diving48, IKEA-ASM, and PennAction. These results demonstrate the effectiveness of incorporating pose cues into VLMs for precise action understanding.

The variety of techniques used in different studies of FER is summarized in Table 6, along with the progression of the techniques of recognition. The table notes the success with which CNN, LSTM, and hybrid models accomplish emotion classification with different datasets, which is attributed to the extraction of spatial and temporal features. Also, the differences in the reported accuracies for different datasets illustrate how the model architecture and the applied preprocessing methods affect model performance.

2.2.5 Supplementary trends in deep learning: summary of key findings

Table 7 below analyzes the performance of DL methods on FER systems. Among others, CK+, JAFFE, FER2013, RAF-DB, and KDEF are used as datasets to train and test emotion recognition systems that usually identify seven different emotions. In contrast, some works aim to identify six or even five emotions. CNNs dominate other architectures with MobileNet (V1, V2), VGG16, ResNet-50, DenseNet-161, and Deep CNNs. Major results for the studies are the following: MobileNet is best suited for real-time applications with high accuracy and efficiency; ResNet-50 and DenseNet-161 greatly outperform other models, and CNNs successfully capture geometric and global features. This strong conclusion is based on diverse and challenging data, including private data, with which many DL models are trained. Architectural changes, such as substituting adaptive

Table 6 Summarization of the diverse deep learning techniques utilized within the FER system

Author with citations	Set of emotions	Techniques used	Dataset	Outcomes
Abir [41]	7	• A batch size of 165 × 165 is set for the CNN input • The structural design is adjusted using the VGG model	KDEF, RAFD, CK+, MUG	The CK+ dataset attains its highest performance at 71.04%
Kim [10]	6	• A CNN extracts spatial features of the demonstrative expression state • An LSTM processes facial expressions, followed by temporal analysis of spatial features	CASME, MMI	The predicted recognition rate using Leave-One-Subject-Out (LOSO) cross-validation is 72.83% for the CASME II dataset and 78.61% for the MMI dataset
Mahoor et al [47]	23 basic and compound emotions	• An LSTM model works in tandem with the ResNet framework to extract the spatial and temporal relations of face areas • This network receives input from facial landmark points	FERA, MMI, DISFA and CK+	The cross-database challenge yields recognition rates of 67.52% for CK+, 54.76% for MMI, 41.93% for FERA, and 40.51% for DISFA
Sang [50]	7	• Data augmentation is a crucial step in DNN • It has been demonstrated that L2 multiclass SVM loss outperforms cross-entropy loss in Facial Expression Recognition (FER)	FERC-2013	Using the BKVGG12 architecture, the FERC-2013 dataset test accuracies for both the public and private tests are given as 71% and 71.9%, respectively
Shabi [49]	7	• Dense SIFT and Regular SIFT are examined and contrasted with the CNN model, and SIFT is integrated with the CNN model • Dense SIFT performs better than SIFT when using CNN	FER 2013 and CK+	The FER dataset reports a classification accuracy of 73.4%, but the CK+ dataset reports a significantly better accuracy of 99.1%
Kahou [45]	7	• Circulating evidence over a sequence by a hidden layer • Accumulation using time-average integration	EmotiW	49.907% is the attained performance level
Graves et al. [48]	6	• The experimental results present two types of LSTM: bi-directional LSTM and uni-directional LSTM • Image sequences exhibit temporal dependencies during the classification process	CK+	Bidirectional and uni-directional LSTMs have classification error rates on the CK+ dataset of 6% and 18.2%, respectively
Mao et al. [158]	Basic facial expressions (various, via AU representation)	• AU-aware Vision Transformer (AU-ViT) with AU branch integration	FERPlus, RAFDB, AffectNet, LSD, and three occlusion test datasets	Superior to state-of-the-art methods across datasets; robust to occlusions
Hosseini et al. [159]	Basic facial expressions (e.g., happy, sad, angry, neutral)	• DL models for facial expression recognition, bias analysis methods	RAF-DB, AffectNet, etc	Found significant accuracy disparities across demographic groups, with up to 15% lower accuracy for some minorities; overall accuracy ranged from 70 to 90%; stressed the need for fairness-aware datasets and models

Table 6 (continued)

Author with citations	Set of emotions	Techniques used	Dataset	Outcomes
Ren, S et al. [160]	Infant facial expressions (e.g., happy, sad, angry, neutral)	• Improved YOLOv8 with transformers, feature extraction, data augmentation	Custom infant facial expression dataset	Achieved high precision infant facial expression recognition with accuracy above 92%; improved detection speed and robustness due to YOLOv8 enhancements and data augmentation
Cruz-Vázquez et al. [161]	3 (Happiness, Sadness, Neutrality)	• CNNs, Fourier Neural Networks (FNNs), Quantum Rotations	Proprietary EEG dataset designed for emotion recognition	Achieved 95% accuracy in emotion classification, with notable performance in distinguishing sad emotions; demonstrated the effectiveness of integrating quantum rotations and FNNs for enhanced feature separability
Yang et al. [162]	Not emotion-focused; focused on human activities under visible light conditions	• Generative Language Model, Vision-Language Alignment, Prompt-based Recognition, Transformer Architecture	VLP-HAR (custom)	Achieved 94.21% accuracy, outperforming CNN-LSTM and Transformer baselines
Zhang, Jiapeng, et al. [164]	Not emotion-focused; focused on fine-grained human actions	• Pose-Enhanced Vision-Language Model (PeVL), Graph-based Pose Encoder, CLIP, Prompt Tuning, Vision-Language Fusion	FineGym, Diving48, IKEA-ASM, PennAction	Achieved state-of-the-art accuracy across datasets; notably improved fine-grained recognition accuracy

pooling and modifying convolutional layers, are cited as possibilities for improving previously stated models' performance.

2.3 Synergistic machine learning and deep learning techniques

Numerous studies have confirmed that wearing a mask affects one's capacity to express and interpret facial emotions. One of the newer studies shows that the concealment of certain lip movements is one of the reasons why it is so difficult to accurately define certain emotional expressions, thus explaining the lack of confidence in such conclusions. Feature extractions are done using DL models VGG-16 and AlexNet. In one study [143], emotion classification is attempted by applying several classifiers following the first two layers (FC6, FC7, FC8). These classifiers are trained with partially obscured facial features. The different research proposed a classification system that combines the displacement recognition of CNN facial motion geometry and the flow recognition of SVM facial motion for facial motion flow classification. This novel approach incorporates both classifiers and provides the highest accuracy for facial expression recognition [144]. Multiple studies have been carried out regarding the functional software designed to understand the relationships between orderly parts of a person's face to get network difference features (NDF). Moreover, an earlier study designed a random forest classifier that classified facial expressions into seven classes, and the results are fairly good [145]. Studies encompassing multiple databases have revealed that combining shallow and deep features has great synergy, surpassing the performance of other existing models [146]. Another approach uses dual-tree m-band wavelet transform with energy entropy measure for high precision feature extraction (DTMBWT) and the Gaussian mixture model (GMM) for classification. This approach examines the characteristics of

Table 7 Comprehensive Review of Deep Learning Approaches for Facial Emotion Recognition (FER)

Reference	Datasets used	Methods/ architectures	No. of emotions detected	Key findings/remarks	Issues and limitations
[115]	CK+, JAFFE, FER2013	MobileNetV2	7	MobileNetV2 demonstrated high accuracy and efficiency in real-time emotion recognition	May struggle with subtle or compound emotions due to lightweight architecture
[116]	FERPlus, RAF-DB	MobileNet	7	MobileNet achieved robust performance with optimized parameters for emotion detection	Limited capacity for deep feature representation; sensitive to noise in complex settings
[142]	JAFFE, CK	CNN	7	CNN-based models effectively classified emotions using geometric and global features	Requires large datasets and careful tuning; prone to overfitting with limited data
[118]	FER-2013, Random Dataset	MobileNet-V1	7	MobileNet-V1 provided a lightweight solution for real-time FER applications	Reduced accuracy on unconstrained datasets and in-the-wild conditions
[119]	FER2013, CK+	VGG16	7	VGG16 achieved high accuracy with adaptive pooling and convolutional layer modifications	High computational demand and memory usage; not ideal for edge deployment
[120]	CK+, RAF-DB, FER-2013	ResNet-50	7	ResNet-50 excelled in feature extraction and emotion classification tasks	Computationally heavy; less suitable for mobile or real-time scenarios
[121]	KDEF	VGG16	6	VGG16 performed well on the KDEF dataset, focusing on six primary emotions	Dataset lacks diversity; model may not generalize well to spontaneous expressions
[122]	CK	ResNet-50	7	ResNet-50 demonstrated state-of-the-art performance on the CK dataset	Performance may degrade on dynamic or occluded facial inputs
[123]	Private VEMO, FER2013	CNN	7	CNN models achieved high accuracy on both public and private datasets	Lack of generalization to diverse environments; private dataset reduces reproducibility
[124]	RAFD, CK+, JAFFE, KDEF, FER2013	DCNN	7	DCNN provided superior performance across multiple datasets with diverse emotions	High training time and parameter complexity; may overfit smaller datasets
[125]	NIST, CMU, Caltech, Faces	FERC	7	FERC models showed high accuracy in recognizing emotions across varied datasets	Scalability and real-time inference remain challenging due to model size
[126]	KDEF, JAFFE	DenseNet-161	7	DenseNet-161 achieved exceptional accuracy on KDEF and JAFFE datasets	Heavy architecture limits deployment on resource-constrained devices
[127]	FER-2013, CK+, FERG, JAFFE	CNN	7	CNN models demonstrated consistent performance across multiple datasets	Requires consistent preprocessing; less robust under real-world noise
[128]	Private Dataset	DCNN	5	DCNN achieved high accuracy on a private dataset with five emotion categories	Limited emotion categories and the use of non-public data affect generalization
[129]	FERC-2013, JAFFE	CNN	7	CNN models performed well on both FERC-2013 and JAFFE datasets	Faces challenges with spontaneous expressions and diverse demographics

expressions at different levels using GLCM features, emotion homogeneity, and contrast indices to effectively identify emotions in images in the database [147]. A new method for predicting emotions has been brought forward recently, which uses a high-level DL approach supported by CNNs for the classification of primary emotions such as happiness or sadness and subsequently extracts secondary emotions from facial images [148]. Another emotion recognition and retrieval system is presented, robust to changes in illumination, occlusion, and posture. This hybrid model tracks the changes in head pose, colour, contrast, and even lighting under static dataset training conditions [149]. A deep geometric feature descriptor is constructed to describe the spatial relations of some facial landmarks, which are grouped into seven categories. Attention-based LSTM (ALSTM) with spatial attention CNN (SACNN) enable combining features for expression recognition accuracy improvement [150]. The GMM-DNN model, for instance, has managed achievements that are by far excellent compared to human raters in their judgments. It has been tested in quiet and noise, in addition to two emotional datasets, and is robust [151]. In ambiguous facial expression scenarios, the proposed model can find the emotion with comparably high accuracy, outperforming or matching existing multimodal emotion recognition systems [152]. Another approach takes a video sequence and averages the segment-level features using deep belief networks (DBN) to form a global video feature. These features are sufficient for a linear SVM to achieve high accuracy in facial expression classification [153].

These studies showcase the utmost improvement in attempts to consolidate geometric, motion, and DL for FER and their accuracy in difficult conditions. Figure 16 describes the algorithmic workflow of FER, which entails various processes such as preprocessing, feature extraction, emotion classification, hyperparameter tuning, and emotion recognition results evaluation. First, the data is prepared through preprocessing, after which ML feature extraction is conducted to extract relevant features from the raw images. Emotion classification uses SVM, Random Forest, Deep Belief Network, Gaussian Mixture, Multilayer Perceptron, and Decision Trees. Following classification, hyperparameter tuning is performed to enhance model performance and the classifiers are trained to increase the accuracy of the models. Finally, an emotion recognition system output is generated to evaluate the results based on common metrics like accuracy, precision, and recall. This diagram demonstrates the combination of preprocessing, feature extraction, and ML model training to attain effective FER systems.

Table 8 gathers information on hybrid learning methods for FER systems, integrating different feature extraction and classification techniques. Numerous research works employ CNNs for feature extraction, commonly used along with SVMs or other classifiers such as Random Forest (RF), Gaussian Mixture Model (GMM), Decision Trees (DT), and LSTM networks. Other methods for feature extraction are Neighborhood Difference Features (NFD), Dual-Tree M-Band Wavelet Transform (DTMBWT), Scale Invariant Feature Transform (SIFT), Deep Belief Networks (DBN), and architecture hybrids such as AlexNet and VGG-16. Among others, CK+, JAFFE, RAF, and SFEW are some of the datasets utilized. The amount of emotions recognized usually ranges from 3 to 7. One of the conclusions derived lauds the effectiveness of combining CNNs with SVMs or LSTMs for accurate emotion classification, the accuracy of some feature extraction methods such as NFD and DTMBWT combined with the right classifiers, and

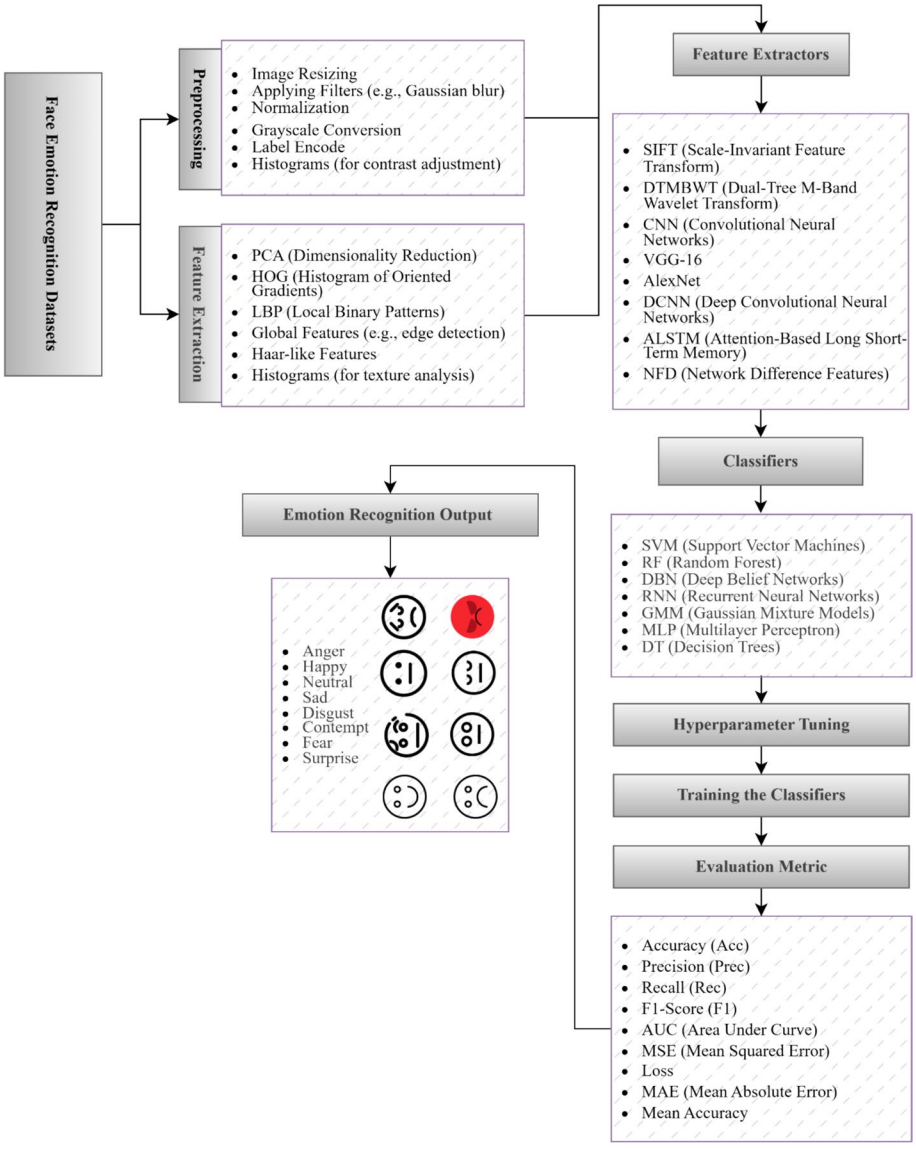


Fig. 16 Comprehensive Workflow for Facial Emotion Recognition Using Hybrid Techniques

the achievement of overall hybrid methodology versus single method approach. More advanced studies claim better results are achieved on dynamic or multisource datasets.

3 Facial expression database

In the domain of facial expression recognition, various databases have been utilized to evaluate system performance. The earliest public facial expression recognition (FER) database emerged in the nineteenth century [51]. Numerous human facial emotions have been explored through 2D facial models; however, challenges such as face aging, illumination variations, and pose changes still persist with this type of database. Subsequently, the introduction of static 3D facial models proved successful. Furthermore, temporal facial databases emerged, encompassing both 2D and 3D face models, thereby addressing the dynamic aspects of facial expressions.

Table 8 Critical Analysis of Hybrid Learning Techniques for Facial Emotion Recognition (FER) Systems

Reference	Datasets used	Feature extraction methods	Classification techniques	No. of emotions detected	Key findings/remarks	Issues and limitations
[143]	M-LFW-FER	AlexNet, VGG-16	SVM	7	Integrating AlexNet and VGG-16 with an SVM classifier yielded high accuracy in emotion recognition	High computational cost; limited performance in low-resource or real-time scenarios
[144]	CK+, BU4D	CNN	SVM	7	Features extracted by a CNN are combined with an SVM for robust emotion classification	May not generalize well to in-the-wild datasets; SVMs are less scalable with large data
[145]	RAF, SFEW	Neighborhood Difference Features (NFD)	Random Forest (RF)	7	Using NFD for feature extraction with an RF classifier achieved high accuracy in recognizing emotions	Sensitive to illumination and occlusion; NFD lacks deep semantic features
[146]	CK+	CNN, Scale-Invariant Feature Transform (SIFT)	SVM	7	Combining SIFT features with CNN-extracted features and an SVM classifier improved emotion detection performance	SIFT is sensitive to noise and scale variation; hybrid extraction increases complexity
[147]	JAFFE	Dual-Tree M-Band Wavelet Transform (DTMBWT)	Gaussian Mixture Model (GMM)	7	Employing DTMBWT for feature extraction and GMM for classification provided precise emotion classification	GMMs assume Gaussian distributions, which may not fit complex real-world data
[148]	JAFFE, FER2013	CNN	CNN	7	A CNN is utilized for both feature extraction and classification, achieving high accuracy in emotion recognition	Overfitting risk on small datasets; lacks temporal modeling for video-based inputs

Table 8 (continued)

Reference	Datasets used	Feature extraction methods	Classification techniques	No. of emotions detected	Key findings/remarks	Issues and limitations
[149]	EMOTIC, FER-13, FER	RNN + CNN	LSTM	7	A hybrid model combining RNN and CNN architectures with LSTM units improved emotion recognition, particularly in dynamic datasets	High training time and parameter count; sensitive to sequence length and quality
[150]	FER2013, CK + , JAFFE	Spatial Attention CNN (SACNN)	Attention-Based LSTM (ALSTM)	7	Combining SACNN and ALSTM enhanced both feature extraction and classification performance in FER	Complex architecture with high computational demands; limited real-time viability
[151]	ESD	Gaussian Mixture Model and DNN (GMM-DNN)	SVM, MLP	6	Integrating GMM-DNN with SVM and Multilayer Perceptron (MLP) classifiers achieved robust performance in emotion recognition	Model complexity and reliance on handcrafted features can limit adaptability
[152]	LUMED-2, DEAP	CNN	Decision Tree (DT)	3	Using a CNN for feature extraction and a Decision Tree for classification provided effective emotion classification for specific datasets	Limited emotion classes; DTs can be unstable and prone to overfitting
[153]	BAUM-1 s, RML, MMI	Deep Belief Network (DBN)	SVM	6	Combining DBN for feature extraction with an SVM classifier demonstrated high accuracy in emotion recognition tasks	DBNs require extensive training time and tuning; may not scale to large datasets
[154]	Multisource	CNN	LSTM	6	Integrating a CNN with LSTM units improved emotion recognition performance across diverse datasets	Performance sensitive to sequence alignment and facial occlusion; computationally intensive

3.1 Extended Cohn – Kanade (CK +)

The CK + dataset [52] encompasses 593 video frames encompassing both posed and spontaneous (non-posed) emotions, comprehensively representing human emotional states. This dataset comprises 2D and 3D images and videos. It is organized based on

the subjects' age, featuring 123 individuals ranging from 18 to 50 years, with 69% being female. Moreover, the dataset includes 13% Afro-American, 91% Euro-American, and other ethnic groups. The emotions within this dataset are classified through action units [63] and are further labelled into seven distinct categories. The images within this dataset possess a pixel resolution of 640×480 and 640×490, with grayscale values represented with an 8-bit precision [13].

3.2 FER 2013

The dataset [53] includes a total of 184 emotions falling under various categories. Each category has various subjects such as gender, age, and ethnicity, making 600 different combinations. The first step is to gather the first 1000 images for every query, which is followed by some post-processing steps to clean the data. After that, the images are sorted into appropriate emotion classes, and those not representative of the class are removed. The final set has roughly 36,000 pictures, organized into eight groups. Seven of these groups are for the basic expressions, while one is kept for neutral labels. Each group has 1000 images except the "disgust" emotion group, which has a total of 547 images [53].

3.3 NovaEmotions

The purpose behind creating the NovaEmotions database [54] is to capture images in real-world settings, encompassing diverse poses and lighting conditions. The images are collected from crowded locations, allowing for the representation of spontaneous emotions across various situational variations using a game algorithm. The database comprises a total of 42,000 images, distributed among ten distinct categories of individuals. The subjects primarily comprise college students or individuals of a young age, falling within the age range of 18 to 25 [54].

3.4 Compound emotions

The CE dataset [7] is a collection comprising 5,060 images, encompassing 22 basic and compound emotions classes. These images pertain to 230 human faces, with a distribution of 100 males and 130 females, and an average age of approximately 23. Subjects within the database are categorized into various ethnicities, including Caucasian, Asian, Hispanic, and African.

3.5 Denver intensity of spontaneous facial action database (DISFA)

DISFA [56] comprises 130,000 clips featuring 27 subjects with a ratio of 12:15 for females and males. The annotation of action units within the video sequences spans a scale from 0 to 5, and this annotation process is performed by two human experts well-versed in the Facial Action Coding System (FACS). Sixty-six facial landmarks accompany each image in the database. The pixel resolution of each facial image is 1024×768.

3.6 Japanese female facial expressions (JAFFE) [58]

Ten Japanese female models were chosen for this project, and they were photographed while posing in the seven fundamental facial expressions, which include six emotive and neutral. In total, they collected 213 images that 60 Japanese participants evaluated according to the six expressions. The resolution of each image is 256×256 pixels.

3.7 Karolinska directed emotional face (KDEF)

The KDEF database [70] encompasses a collection of 4,900 images capturing human facial emotion . These images are organized into seventy separate images, each depicting seven distinct expressions. The pictures were captured from five different angles. The fundamental pixel resolution for each facial image within the database is 562×762.

3.8 MMI

The dataset MMI [71] features a collection of video and still images encompassing 75 subjects, totalling 2900 frames. Action Units are shown in the video AUs and are partially captured at the image level, which is the frame level. An Action Unit may be present at an onset, apex, neutral, or offset level. The set features a total of 238 video images which are split into 28 categories for both females and males. The original images of faces in the dataset have pixels of the dimension 720 by 576.

3.9 Binghamton university 3D facial expression (BU-3DFE)

The BU's main focus is to create a 3D facial emotion database BU-3DFE [58], hoping to enhance knowledge of human behaviour. This database has 100 participants, 56 females and 44 males, demonstrating 6 different expressions. This database carries 25 3D facial expressions per subject, each model with 83 manually annotated facial landmarks. The original images of faces in this set have a resolution of 1040 by 1329 pixels.

3.10 Self-Created databases

Besides the public databases already cited, some researchers used real-time data of facial expressions captured by end users for both more challenging and less fulfilling parts of the evaluation. This data is captured using MATLAB-integrated cameras or NIR, as well as thermal and Kinect cameras.

In the work described in the paper [3], 3D face modelling is done with a Microsoft Kinect. This method tried to solve problems accompanying head and shoulder posture changes, illumination changes, and lighting-produced issues that affect the quality of impression recognition systems, primarily based on 2D picture images; this is how the Microsoft Kinect works. The device is relatively low in price and easy to use but also quite low in resolution. However, it achieves a high photograph capture rate of 30 frames per second. An infrared emitter and two cameras are used. One is a color camera, and the other is an infrared camera designed to capture depth information. Considering the variety of sensors used in FER systems, including those different from cameras, it is important to point out that using only one sensor to improve recognition performance poses some limitations. These limitations can be improved when a variety of sensors are employed with FER systems.

Table 9 shows the overall summary of the FER datasets, including details, resolution, expression type and online available links.

4 Addressing challenges and evaluating performance in facial expression recognition

Facial expression recognition systems face several challenges, including variations in lighting, occlusions, differences in facial structures, and real-time processing demands, which can affect their accuracy and robustness. To address these challenges, researchers

Table 9 Overview of FER based databases

Database name	Year	Image/video count & subject details	Resolution	Expression type	Link
CK + [52]	2010	• 593 video frames, 123 subjects (ages 18–50), posed & natural expressions	640 × 490 & 640 × 480	Posed & Spontaneous	https://www.kaggle.com/datasets/shuvoalok/ck-dataset
FER2013 [53]	2013	• 36,000 images, 7 basic expressions + neutral, varying distribution per class	48 × 48	Posed	https://www.kaggle.com/datasets/msambare/fer2013
Nova Emotions [54]	2016	• 42,000 images, 10 person types, collected in crowded environments	Variable	Posed and Spontaneous	https://www.kaggle.com/datasets/ananthu017/emotion-detection-fer
CE [6]	2014	• 230 subjects (100 male, 130 female, avg. age 23), 22 basic & compound emotion classes	3000 × 4000	Posed & Induced	S. Du, Y. Tao, & A.M. Martinez , Compound facial expressions of emotion, Proc. Natl. Acad. Sci. U.S.A. 111 (15) E1454-E1462, https://doi.org/10.1073/pnas.1322355111 (2014)
DISFA [56]	2013	• 130,000 video sequences, 27 subjects (12 female, 15 male)	1024 × 768	Spontaneous	https://www.mohammadmahoor.com/pages/databases/disfa/
JAFFE [57]	1998	• 213 images, 10 female Japanese models	256 × 256	Posed	http://www.kasrl.org/jaffe.html
KDEF [70]	1998	• 4900 images, 7 expressions, 5 angles per expression	562 × 762	Posed	https://www.kaggle.com/datasets/tom99763/testtt
MMI [71]	2002	• 2900 video frames & images, 75 subjects, 238 video frames across 28 categories	720 × 576	Posed, Spontaneous, Audio-Visual	https://opendatalab.com/OpenDataLab/MMI

employ advanced preprocessing techniques, such as illumination correction and occlusion handling, alongside powerful DL models like CNN with attention mechanisms to better capture subtle and dynamic facial cues. Performance evaluation plays a crucial role in validating these systems, using metrics such as accuracy, precision, recall, F1-score, and confusion matrices to measure effectiveness. Additionally, testing on diverse datasets and considering real-time factors such as speed and latency help ensure models perform well across various conditions and applications. The subsequent subsections carry an overall discussion on the challenges and performance evaluation.

4.1 Challenges in facial expression recognition

Facial Expression Recognition (FER) systems are inherently complex due to the dynamic nature of human emotions and the variability in facial appearances. One of the primary technical challenges is illumination variation. Faces captured under different lighting conditions can appear drastically different, affecting the visibility of crucial facial features used for expression analysis. Uneven lighting can create shadows, highlights, or low-contrast regions that obscure subtle muscle movements critical for distinguishing between expressions. Conventional preprocessing techniques such as histogram

equalization or gamma correction help normalize brightness and contrast but often fail in highly uncontrolled environments. More sophisticated approaches, such as Retinex-based illumination normalization or DL-based image enhancement, aim to compensate for lighting variability. Still, their effectiveness depends on the diversity and quality of training data. Illumination changes remain a core obstacle, especially for real-world applications like surveillance or human–computer interaction, where lighting is unpredictable.

Closely related to illumination issues are pose variations and head movements, which introduce another layer of complexity. FER systems generally assume a frontal or near-frontal face; subjects often move their heads in practical scenarios, resulting in non-frontal views. These pose variations distort the spatial relationships between facial landmarks and can hide or deform key regions such as the eyes, mouth, or eyebrows. This misalignment complicates feature extraction and expression classification. Researchers use methods like 3D face modeling, pose normalization, or multi-view training to address this problem. 3D morphable models reconstruct a frontal view from arbitrary poses, while multi-view DL models train on datasets containing faces at various angles to increase robustness. Despite these advances, accurately recognizing expressions from extreme angles or fast head movements remains challenging.

Occlusion is another major hurdle in FER, where parts of the face may be hidden by accessories (glasses, masks), hands, hair, or environmental objects. Occlusion causes partial or complete loss of critical facial information, making it difficult for models to infer the underlying expression correctly. Technically, occlusion leads to missing data or noisy inputs, which traditional handcrafted features and early ML classifiers handle poorly. Modern approaches employ robust facial landmark detection algorithms capable of identifying visible key points despite occlusions, and DNNs with spatial attention mechanisms that focus on unoccluded regions. Generative adversarial networks (GANs) have also been explored to reconstruct occluded facial areas before expression recognition. Still, these methods require extensive training and are sensitive to the quality of generated outputs. Occlusion remains especially problematic in real-world contexts, such as masked faces during pandemics or occlusions in crowded environments.

Another challenge is the inter-subject variability inherent in facial expressions. People express emotions differently based on individual facial anatomy, age, gender, ethnicity, and cultural background. For example, a smile can vary from a subtle lip movement to a broad grin with teeth exposed, and some cultures may express emotions with different facial muscle patterns or intensities. This variability complicates the generalization ability of FER models, which often overfit specific demographics present in training datasets. To counter this, researchers emphasize the need for large, diverse, and balanced datasets that capture a wide spectrum of expressions from varied populations. Transfer learning and domain adaptation techniques are also used to improve model robustness across different demographic groups. However, acquiring and annotating such rich datasets remains costly and labour-intensive, limiting widespread implementation.

Expression intensity and subtlety introduce further technical challenges. Facial expressions range from overt displays to subtle micro-expressions that last only fractions of a second but convey important emotional cues. Detecting and distinguishing these low-intensity or fleeting expressions requires highly sensitive models with fine temporal resolution. While CNNs excel at spatial feature extraction, capturing temporal

dynamics necessitates recurrent neural networks (RNNs), long short-term memory networks (LSTMs), or temporal convolutional networks (TCNs) that can model changes over time. Moreover, datasets that include labeled micro-expressions are scarce, making it difficult to train and validate such temporal models. Real-time FER applications add pressure to balance accuracy with processing speed, as temporal models are often computationally expensive and prone to latency.

Finally, evaluation and benchmarking FER systems pose their own set of challenges. The performance of FER models is commonly assessed using metrics such as accuracy, precision, recall, F1-score, and confusion matrices on benchmark datasets. However, these datasets often have limited diversity and may not represent real-world conditions like varying illumination, occlusion, or spontaneous expressions. This leads to inflated performance results that do not always translate to practical scenarios. Cross-dataset evaluation and real-time testing are necessary to understand true model robustness. Additionally, the absence of standardized protocols for FER evaluation complicates comparing different methods. Addressing these evaluation challenges requires developing more comprehensive datasets, realistic testing environments, and standardized benchmarking protocols to ensure reliable and generalizable FER models.

Beyond the primary challenges already discussed, several other technical and practical issues impact the development and deployment of Facial Expression Recognition systems. One such challenge is the ambiguity and overlap between facial expressions, where certain emotions share similar facial cues, making it difficult to classify them distinctly. For example, expressions of fear and surprise can be easily confused due to shared features like widened eyes and raised eyebrows. This ambiguity necessitates more sophisticated models capable of capturing subtle differences or leveraging contextual information. Another less addressed challenge is the temporal inconsistency in spontaneous expressions—unlike posed expressions, natural emotions often evolve gradually and may co-occur with other facial movements like speech or head gestures, complicating accurate recognition. Furthermore, privacy and ethical concerns arise when deploying FER technology, especially in public or sensitive environments, requiring careful consideration of data protection, consent, and potential misuse. Finally, the computational cost and energy efficiency of FER systems, especially those relying on DL models, pose practical constraints for deployment on edge devices like smartphones or wearable technology, where resources are limited. Addressing these diverse and emerging challenges remains critical for advancing FER towards more reliable, ethical, and user-friendly applications.

Table 10 provides a comparative evaluation of the primary challenges encountered in Facial Expression Recognition (FER) systems, highlighting their technical impacts, common approaches to mitigation, and the inherent limitations of these solutions. Illumination variation and pose differences significantly distort facial features, necessitating sophisticated preprocessing and modeling techniques such as normalization and 3D pose correction. Occlusions and inter-subject variability introduce missing data and expression diversity, addressed through robust landmark detection and diverse training datasets. Subtlety in expression intensity demands temporal modeling to capture fleeting micro-expressions, though limited annotated data presents a bottleneck. Evaluation challenges underscore the need for standardized benchmarks and realistic testing to ensure generalizability. Additional concerns, such as ambiguity between expressions, temporal inconsistencies, privacy issues, and computational constraints, further

Table 10 Comparative Analysis of Challenges and Mitigation Strategies in Facial Expression Recognition

Challenge	Technical impact	Common solutions/approaches	Limitations
Illumination Variation	Alters facial feature visibility, affects contrast and texture	Illumination normalization (e.g., histogram equalization, Retinex), data augmentation, DL-based enhancement	Limited effectiveness in highly uncontrolled lighting
Pose Variation	Misalignment of facial landmarks, occlusion of key facial regions	3D face modeling, pose normalization, multi-view training	High computational cost; struggles with extreme poses
Occlusion	Partial or full loss of facial information	Robust landmark detection, spatial attention in CNNs, GAN-based reconstruction	Requires complex models; sensitive to occlusion extent
Inter-subject Variability	Differences in facial anatomy, culture, age, gender	Diverse datasets, transfer learning, domain adaptation	Data collection is costly; potential demographic bias
Expression Intensity & Subtlety	Difficulty detecting micro-expressions and subtle changes	Temporal models (RNNs, LSTMs, TCNs), high-resolution datasets	Scarcity of labeled data; computationally expensive
Evaluation & Benchmarking	Overfitting to limited datasets, lack of real-world representativeness	Cross-dataset testing, real-time evaluation, standardized protocols	Lack of universally accepted benchmarks
Ambiguity Between Expressions	Overlapping facial features between emotions	Context-aware models, multimodal fusion (audio, physiological signals)	Complexity in modeling context; multimodal data needed
Temporal Inconsistency	Spontaneous expressions evolve gradually, mixed with other movements	Dynamic temporal modeling, continuous learning	High model complexity; data annotation challenges
Privacy & Ethical Concerns	Potential misuse, data protection, user consent	Privacy-preserving ML, anonymization, ethical guidelines	Regulatory and technical challenges
Computational Cost	High resource demands limit edge deployment	Model compression, lightweight architectures, hardware acceleration	Trade-off between accuracy and efficiency

complicate FER deployment. Despite advances in DL and data augmentation, these challenges remain active research areas, as no single solution fully resolves all limitations. The table succinctly encapsulates these complex trade-offs, providing a framework for understanding and addressing the multifaceted nature of FER challenges.

4.2 Evaluation metrics

Metrics for evaluation are essential for evaluating and measuring the FER system's performance. To help with metrics detection and binary classification goals, several terms, such True Positive (TP), False Positive (FP), True Negative (TN), and False Negative (FN), are used in computations. The four ways that these terminologies are classified under use precision, recall, accuracy, and F1-score, each of which employs unique features [59].

- *Precision (P_C)* It is a method for estimating the percentage of pertinent outcomes that are accurately identified in the output window and for quantifying accuracy [60].

$$\mathrm{P_C} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{23}$$

- *Recall (RC)* It is calculated as the ratio, also known as sensitivity, between the number of pertinent emotional responses and the total number of emotionally charged pictures [61].

$$\mathrm{R_C} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{24}$$

- *Accuracy (AR_C)* It is applied to calculate the proportion of both (TP and TN) cases among all the cases that are analyzed.

$$\mathrm{AR_C} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN}} \tag{25}$$

- *F1-Score* Two metrics are distinguished by their utilization of either spatial or temporal data. The term "frame-based F1-score" or F1-frame refers to the first one, and "event-based F1-score" or F1-event to the second. These metrics capture distinct characteristics of the outcomes. In this discussion, we focus on F1-frame, which exhibits predictive capability concerning spatial stability.

$$\mathrm{F_1} - \mathrm{frame} = \frac{2\mathrm{R_C} * \mathrm{P_C}}{\mathrm{R_C} + \mathrm{P_C}} \tag{26}$$

- *F1-event* Since the expression follows a temporal signal, the event-based F1-score is employed to calculate the expression recognition performance at the segment level. It possesses the predictive potential of temporal data [62].

$$\mathrm{F_1} - \mathrm{event} = \frac{2\mathrm{ER_C} \times \mathrm{EP_C}}{\mathrm{R_C} + \mathrm{P_C}} \tag{27}$$

- *Matthews Co-relation Coefficient (MCC)* [63] It assesses all detection categories and produces a value within the range of 0 to 1, indicating a complete lack of agreement. A score of 0 would imply random prediction, and thus both variables can be considered independent. This offers a more equitable assessment of predictions compared to previous metrics, even though obtaining all four detection cases (TP, FP, FN, TN) might not be feasible. It can be computed using:

$$\mathrm{MCC} = \frac{\mathrm{TP_C} \times \mathrm{TN} - \mathrm{FP_C} \times \mathrm{FN}}{\sqrt{(\mathrm{TP_C} + \mathrm{FP_C})\,(\mathrm{TP_C} + \mathrm{FN})\,(\mathrm{TN} + \mathrm{FP_C})\,(\mathrm{TN} + \mathrm{FN})}} \tag{28}$$

5 Conclusion

This paper thoroughly investigates facial expression analysis using both DL and traditional methods. Additionally, it provides a succinct synopsis of face action units [63]. The study concludes that the DL methodology performs better when managing large databases than the conventional way. To improve accuracy and maximize feature vectors, this survey study addresses a variety of pre-processing strategies, feature extraction approaches, and feature descriptor modifications. A table summarizing the feature reduction technique and the conventional approach's facial expression categorization is also included. Certain limitations that require addressing in future research are highlighted:

- Enhancing accuracy in identifying facial emotions accurately.
- Addressing challenges related to obstructed facial databases.

- Improving the evaluation metrics for applications based on temporal images or videos.

The paper introduces the DL-based approach to game engine facial expression recognition (FER) and examines the five constituent layers of a convolutional neural network (CNN). As CNN-based FER is focused almost exclusively on spatial information, it is not well suited for temporal changes in facial features. To solve this problem, the hybrid architecture CNN-LSTM (RNN) is proposed to deal with the temporal data.

Various CNNs like LeNet5, ZFNet, and others are available in the CNN-based FER approach, while ResNet (Residual Network) has gained prominence in numerous computer vision applications. A DNN is employed for relative learning, predicting emotions, and online reinforcement learning, which is particularly advantageous when dealing with substantial databases due to its efficiency. Following this, a review of related work in FER based on DL approaches is presented in a table. However, certain challenges persist that require attention in future research:

- The DL approach demands significant space for training and testing datasets, which can be time-consuming and complex when deployed on mobile platforms.
- Proficiency in working with DL (CNN) is necessary, as many users lack the knowledge of its workings and integration possibilities with various applications.

Numerous feature extraction techniques are utilized in the conventional approach, including HOG, LBP, and others. However, these methods tend to be time-consuming, necessitating the application of dimensionality reduction techniques to trim down feature vectors. The employment of optimization algorithms becomes essential to fine-tune results. Despite these efforts, these techniques often fail to achieve real-time recognition for 3D facial emotion images. Conversely, the DL-based approach demands a sizable dataset for effective emotion recognition. This review paper delves into approximately 25 research papers encompassing both approaches. It explores fundamental traditional methods and establishes a foundational understanding of key terms pertinent to the DL methodology.

In addition to theoretical advancements, the practical deployment of FER systems requires a focus on real-world generalizability, efficient processing under constrained conditions, and privacy-aware design. Researchers are encouraged to prioritize cross-domain validation, the use of diverse and multimodal datasets (e.g., combining facial, audio, and physiological signals), and the development of interpretable models to ensure responsible deployment.

For practitioners, adopting FER solutions that align with application-specific needs—such as real-time feedback in online education or non-intrusive affect detection in healthcare—will be essential. Recommendations include integrating FER models with lightweight hardware, implementing adaptive learning capabilities, and ensuring compliance with ethical standards.

Furthermore, collaborative efforts between academia and industry can play a transformative role in overcoming current limitations. These partnerships can enable access to large-scale, diverse datasets, foster user-centered design practices, and accelerate the translation of cutting-edge research into usable, context-aware systems. Joint initiatives can also address regulatory and deployment challenges, advancing FER technologies toward broader societal impact.

5.1 Future directions

The recognition of automated facial expression analysis has been accomplished substantially with the integration of these two approaches. However, as delineated in the conclusion section, certain challenges need to be addressed in forthcoming research endeavours. Expanding Motion Descriptors is necessary to diversify descriptors to suit applications featuring varying video/image sequence durations. Challenges such as occlusions in facial alignment, background colours, and environmental conditions persist during image acquisition, leading to compromised classification accuracy and performance. Consequently, this critical issue demands attention in future research. Capturing images under varying environmental conditions remains a formidable challenge that requires addressing in the future.

Moreover, the performance of AFER systems on 3D facial data and frontal facial images has historically been limited, highlighting the need for novel approaches to accurately capture subtle facial movements and expressions in these complex scenarios. In this context, exploring hybrid models that synergistically combine the strengths of conventional algorithms and DNNs, such as CNN-LSTM architectures, offers a promising pathway to improve both spatial and temporal feature learning while maintaining computational efficiency. Another significant avenue for future work lies in tailoring facial expression recognition (FER) systems to domain-specific applications. For instance, recognizing students' facial expressions in online education environments remains an open challenge, where enhanced preprocessing steps combined with DL inference can significantly boost accuracy and practical usability.

In addition, integrating auxiliary models and adaptive learning capabilities can facilitate real-time FER deployment in diverse fields, including gaming, deception detection, automated driving, and marketing research. Such integration can improve system responsiveness and adaptability to new or changing contexts. Furthermore, advancing cross-domain validation strategies using diverse, multimodal datasets, such as facial, audio, and physiological signals, will ensure FER models generalize well across different populations and environments. To accelerate FER technologies' development and ethical deployment, fostering collaborative efforts between academia and industry is imperative. These partnerships can provide access to large-scale, heterogeneous datasets and promote the design of privacy-aware, interpretable, and ethically sound FER systems. Ultimately, these concerted research efforts will help translate FER technologies into practical, reliable, and user-friendly applications with broad societal impact.

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